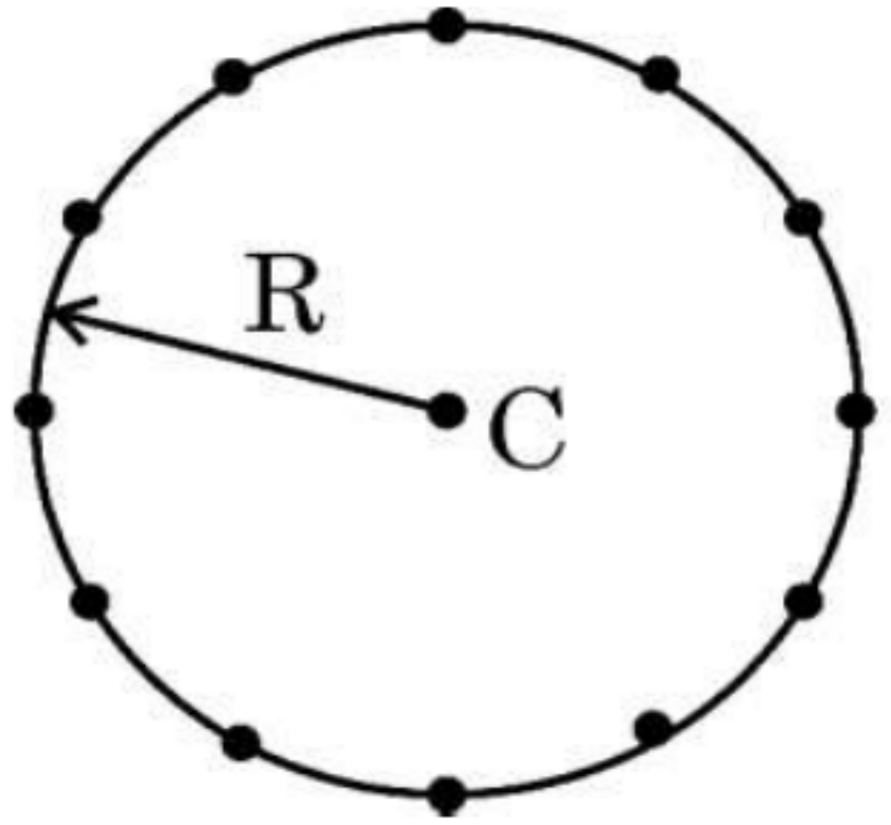
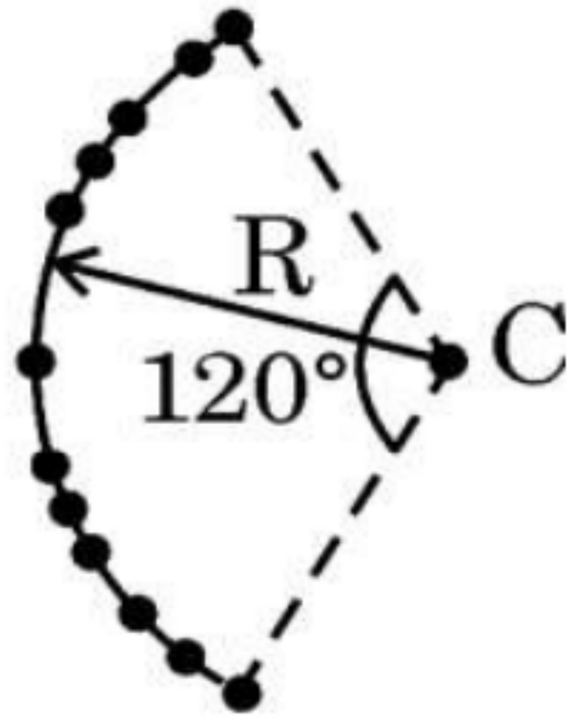
26. (a) Twelve negative charges of same magnitude are equally spaced and fixed on the circumference of a circle of radius R as shown in Fig. (i). Relative to potential being zero at infinity, find the electric potential and electric field at the centre C of the circle.

(b) If the charges are unequally spaced and fixed on an arc of 120° of radius R as shown in Fig. (ii), find electric potential at the centre C. *3*



(i)



(ii)

a)

$V_C = V_1 + V_2 + V_3 + \cdots + V_{12}$

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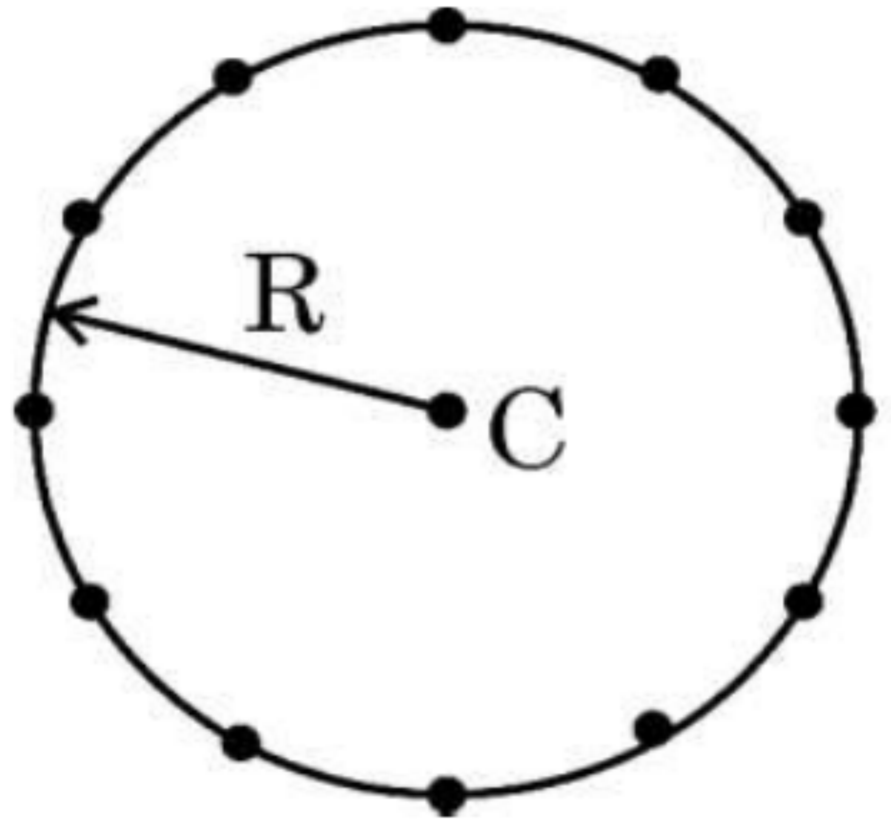

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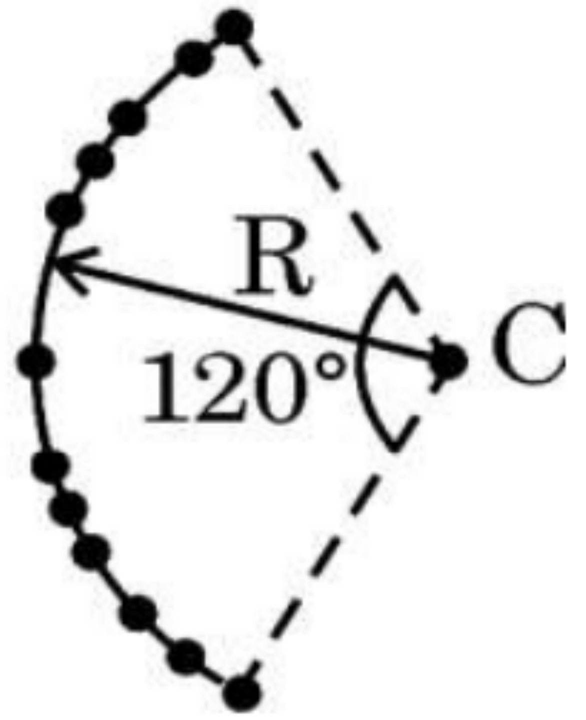

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Since all twelve charges are same $(-q)$ and are at equal distance (R) from the centre (C).

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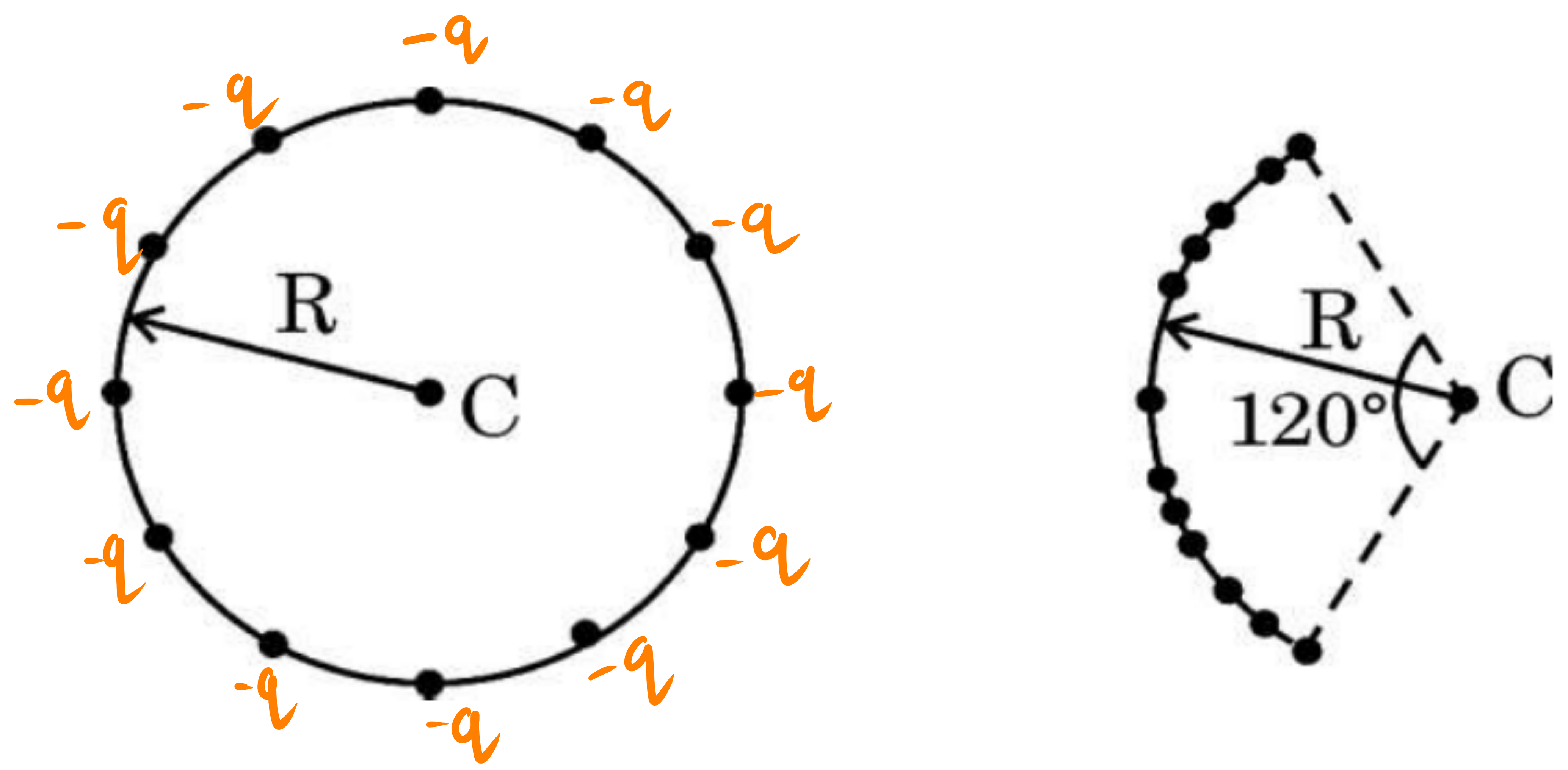

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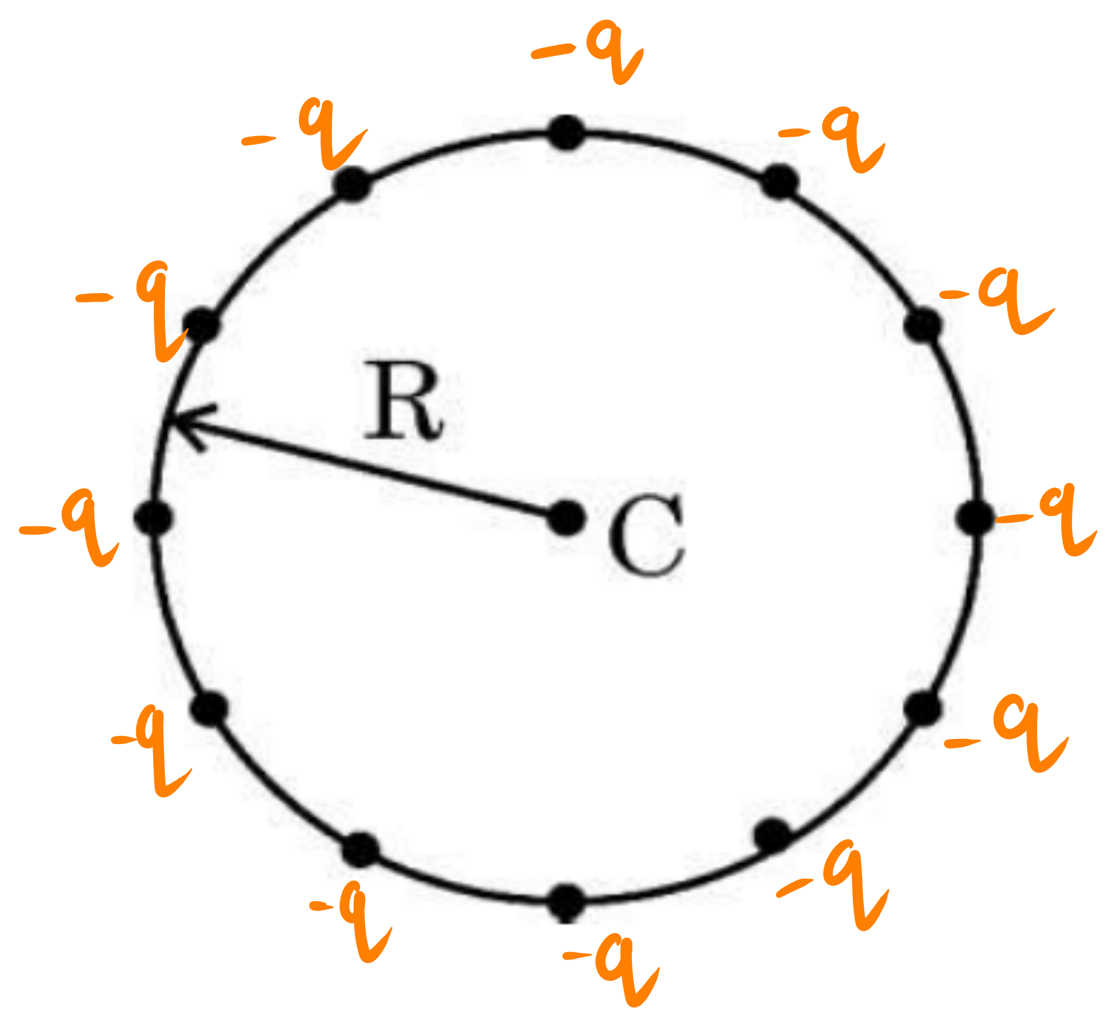
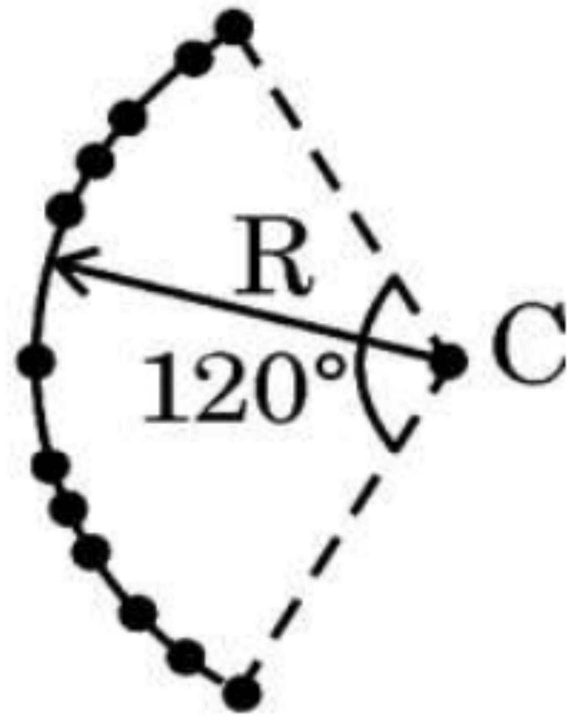


(i)



(ii)

a)

$V_C = V_1 + V_2 + V_3 + \cdots + V_{12}$

Since all twelve charges are same (-q) and are at equal distance (R) from the centre (C).

$V_1 = V_2 = \cdots = V_{12} = V = \frac{1}{4\pi\varepsilon_0}\frac{-q}{R}$

$V_C = 12V$

$$\boxed{V_C = 12 \times \frac{1}{4\pi\varepsilon_0}\frac{-q}{R}}$$

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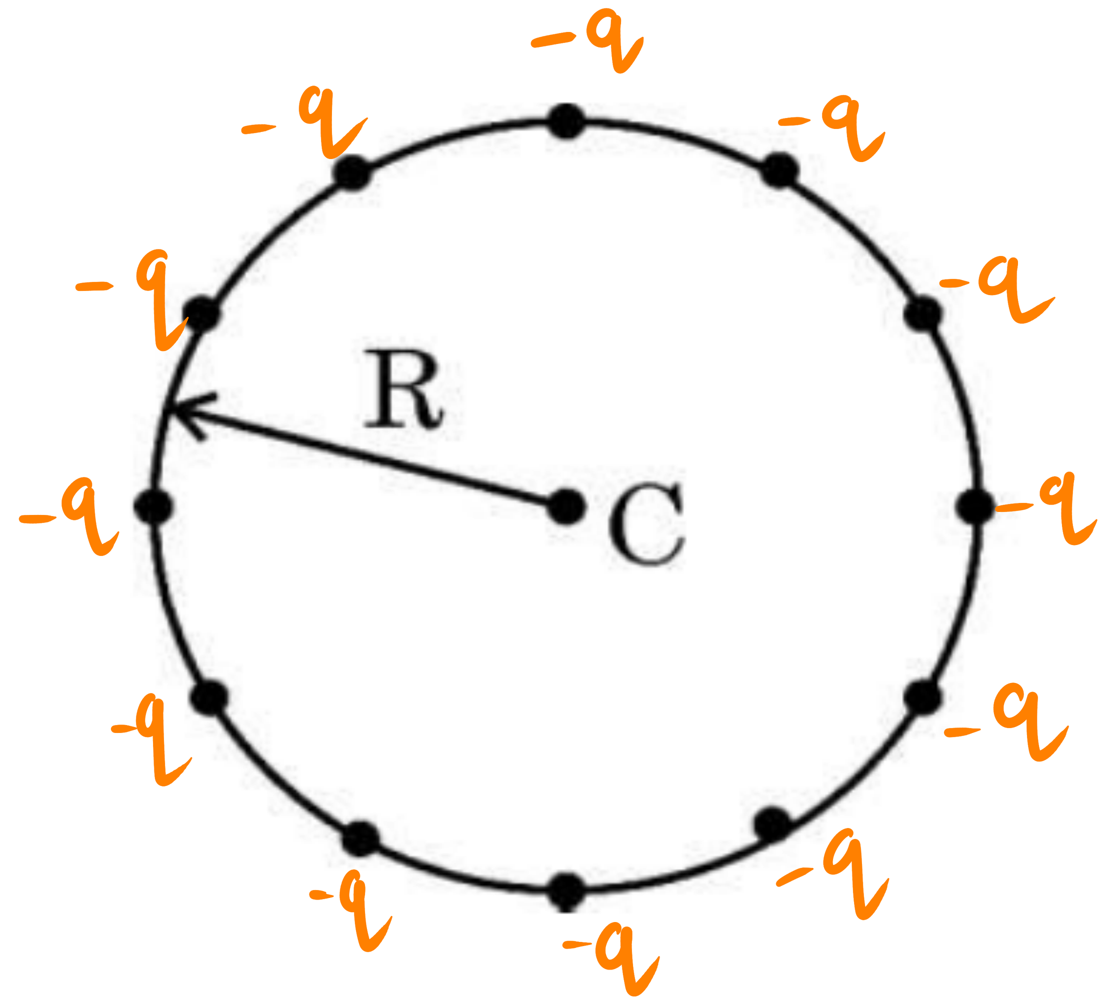
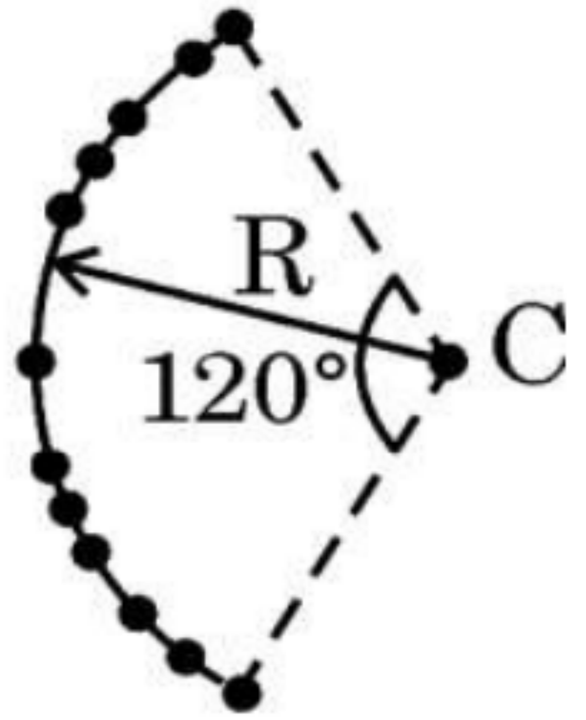


(i)



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$V_c = 12V$

$$\boxed{V_c = 12 \times \frac{1}{4\pi\varepsilon_0}\frac{-q}{R}}$$

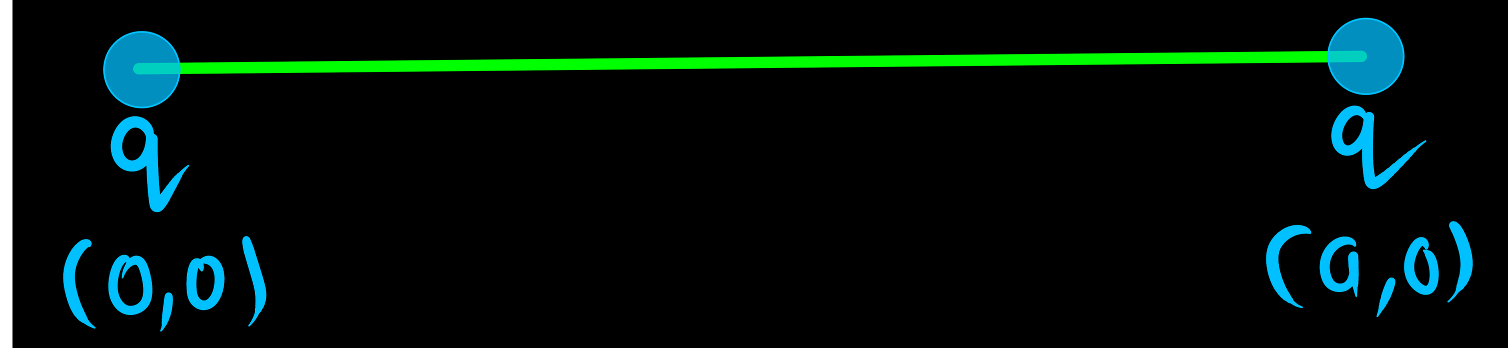
b) Since, potential is a scalar quantity. Similar to part (a)

$V_c = V_1 + V_2 + \cdots + V_{12}$

$V_c = 12V \quad \left(\because V_1 = V_2 = \cdots = V_{12} = V = \frac{1}{4\pi\varepsilon_0}\frac{-q}{R}\right)$

$$\boxed{V_c = 12 \times \frac{1}{4\pi\varepsilon_0}\frac{-q}{R}}$$

(b) (i) Consider two identical point charges located at points (0, 0) and (a, 0).



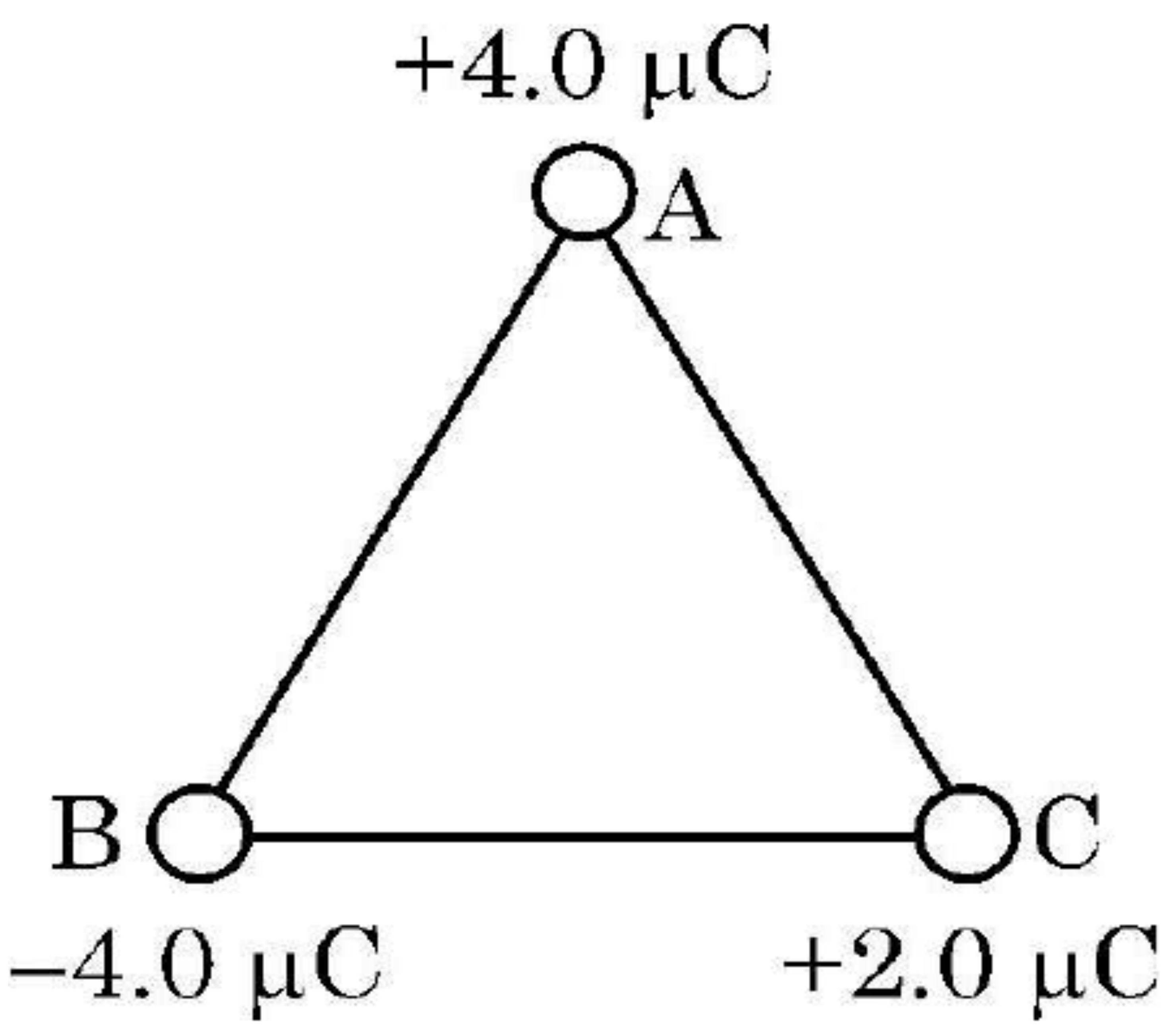
(1) Is there a point on the line joining them at which the electric field is zero ?

(2) Is there a point on the line joining them at which the electric potential is zero ?

Justify your answers for each case.

(ii) State the significance of negative value of electrostatic potential energy of a system of charges.

Three charges are placed at the corners of an equilateral triangle ABC of side 2.0 m as shown in figure. Calculate the electric potential energy of the system of three charges.



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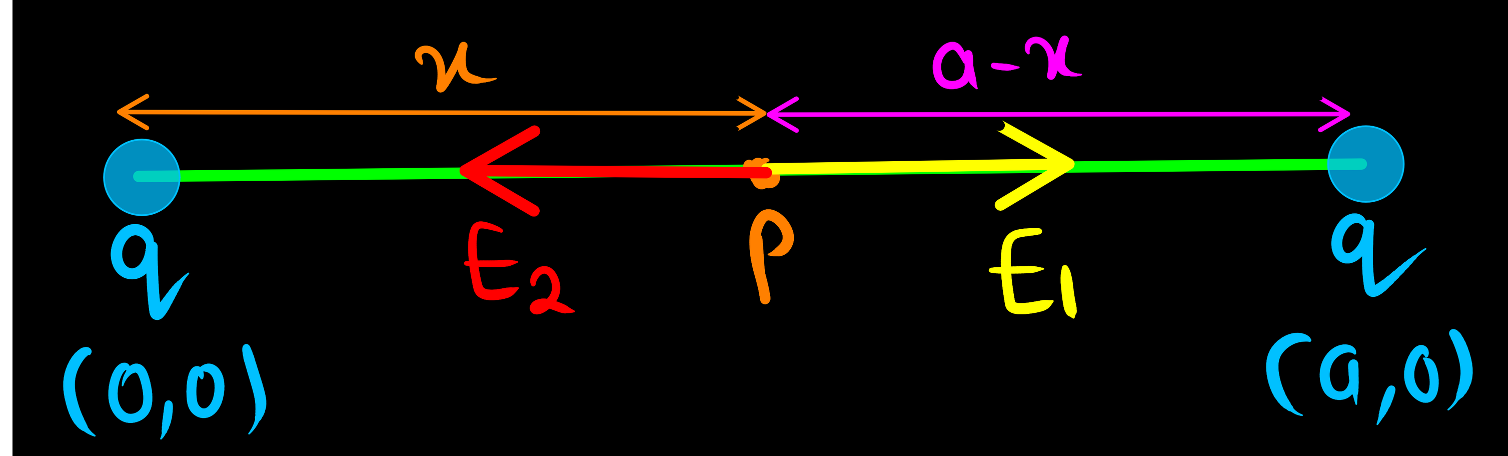

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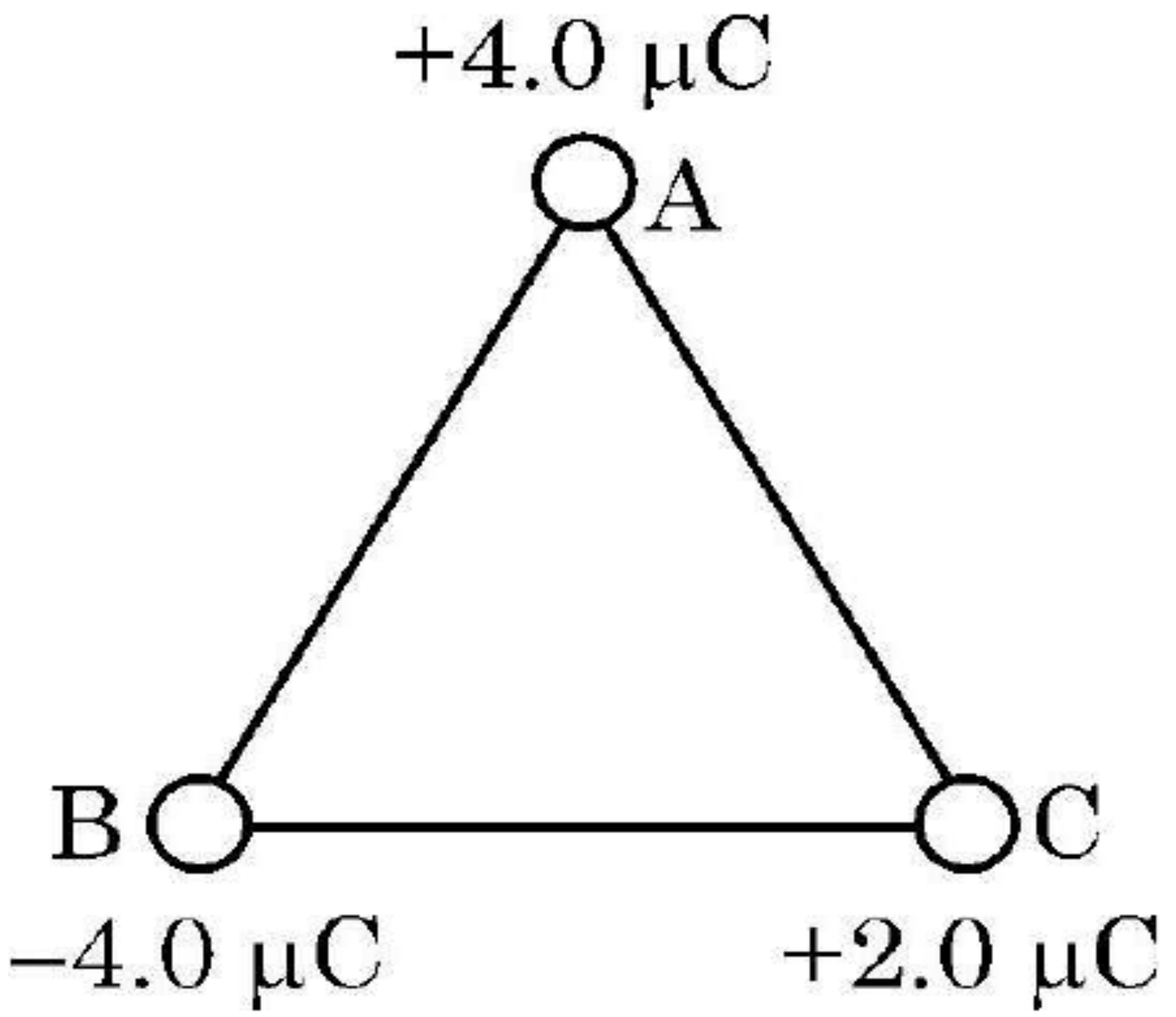

(i) (1) Let at point P, Electric field is zero

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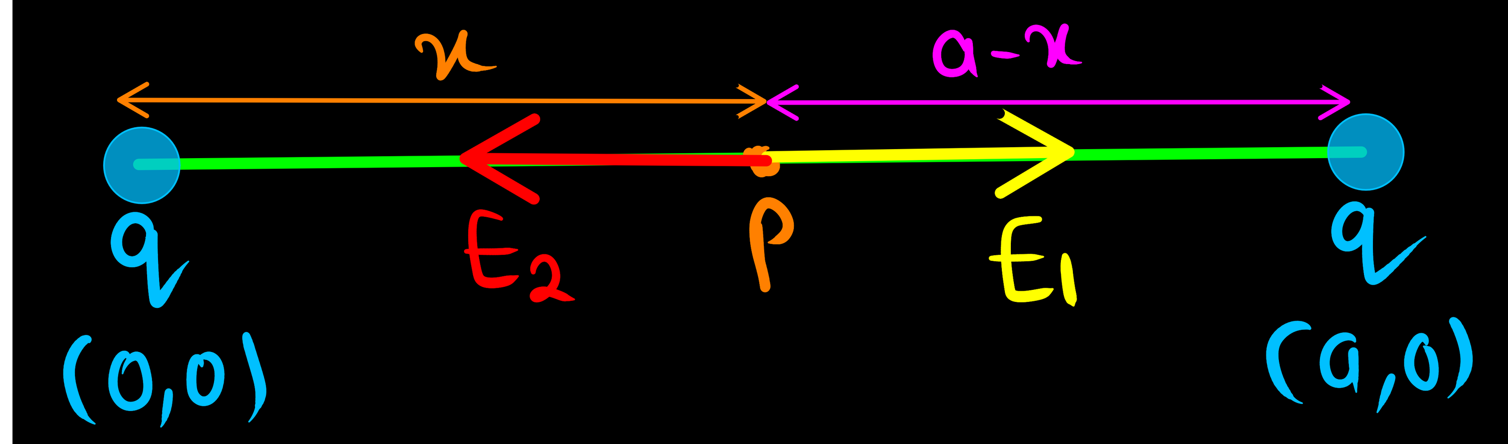

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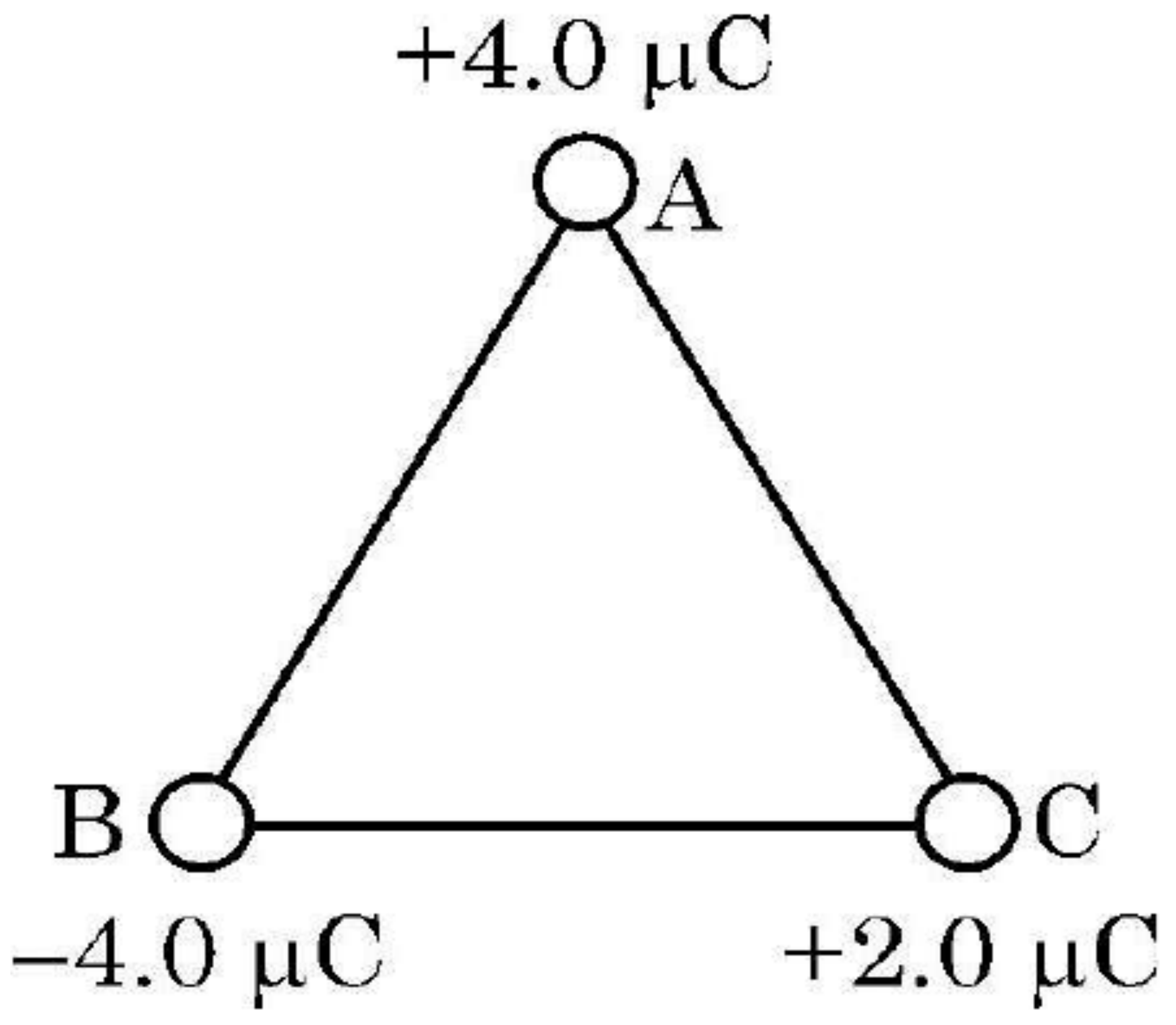

(i) (1) Let at point P, Electric field is zero

$$|\vec{E}_1| = |\vec{E}_2|$$

$$\cancel{\frac{1}{4\pi\varepsilon_0}} \frac{\cancel{q}}{x^2} = \cancel{\frac{1}{4\pi\varepsilon_0}} \frac{\cancel{q}}{(a-x)^2}$$

$$(a-x)^2 = x^2$$

$$a - x = x$$

$$a = 2x$$

$$\boxed{x = \frac{a}{2}}$$

$\therefore$ at the centre $(\frac{a}{2}, 0)$ of the line joining the two charge Electric field will be zero.

(b) (i) Consider two identical point charges located at points (0, 0) and (a, 0).

(1) Is there a point on the line joining them at which the electric field is zero ?

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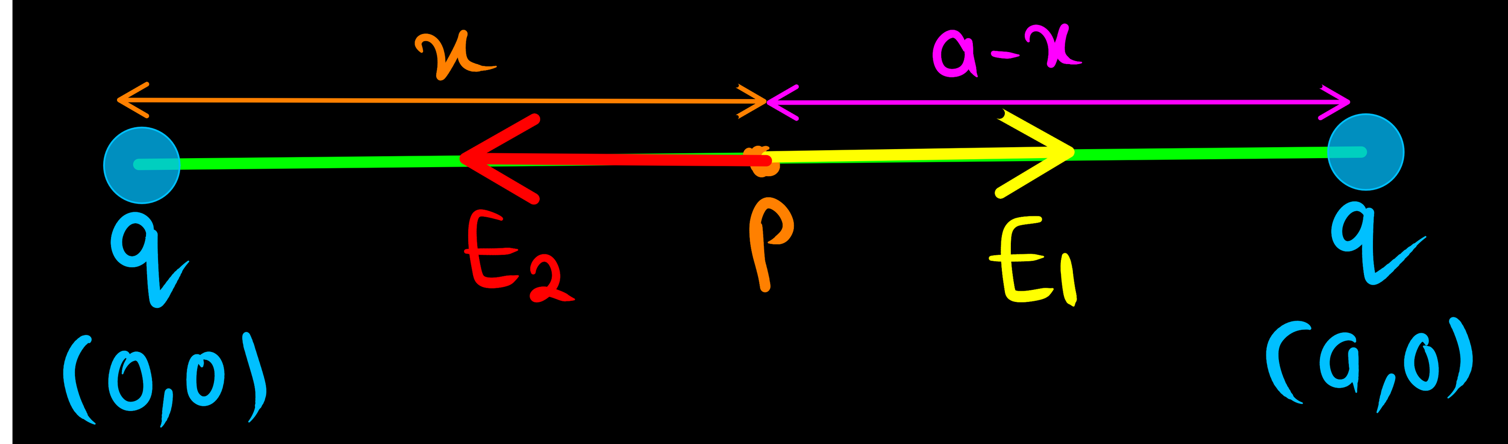
Justify your answers for each case.



(ii) State the significance of negative value of electrostatic potential energy of a system of charges.

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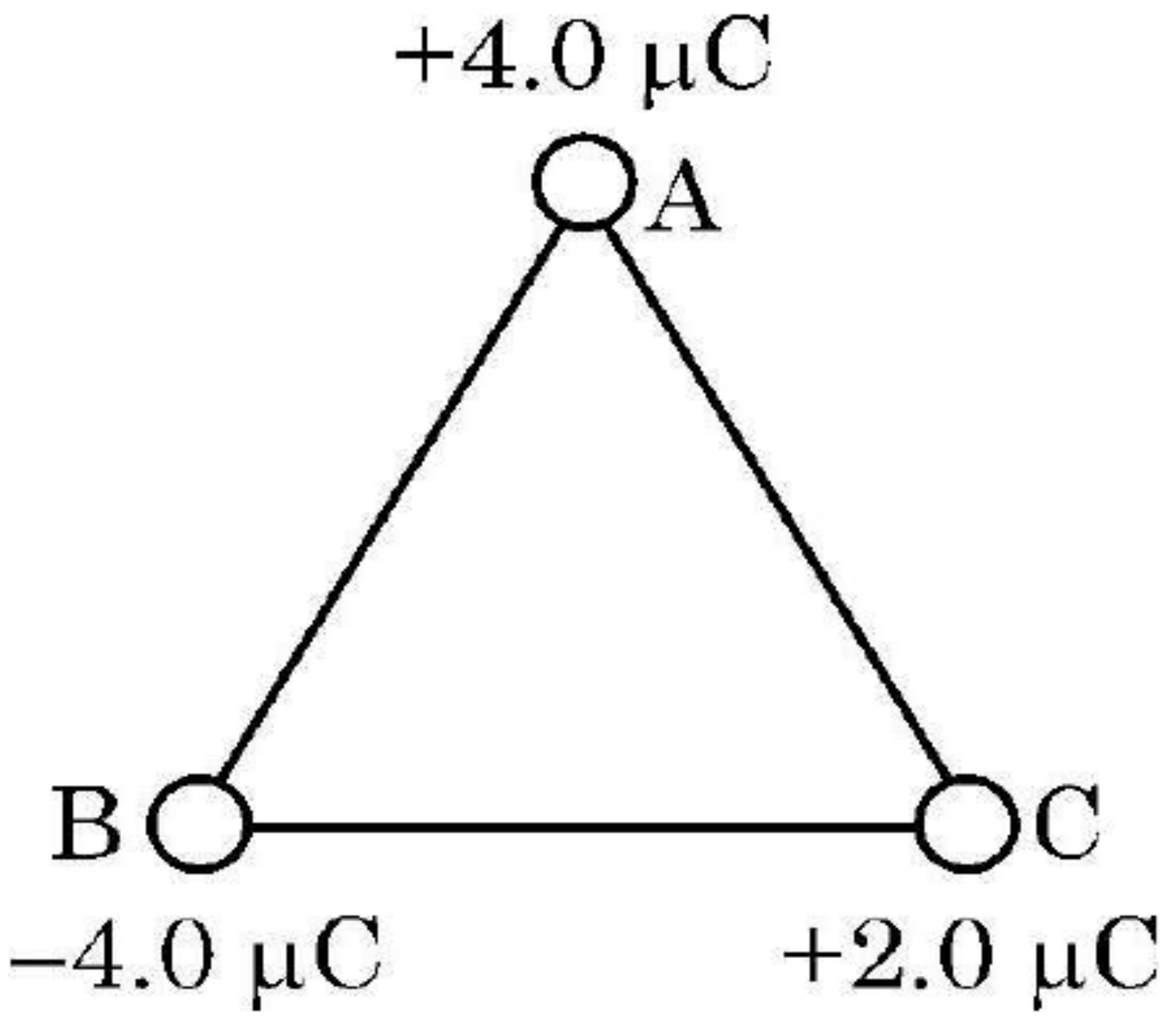

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$|\vec{E}_1| = |\vec{E}_2|$

$\cancel{\frac{1}{4\pi\varepsilon_0}} \frac{\cancel{q}}{x^2} = \cancel{\frac{1}{4\pi\varepsilon_0}} \frac{\cancel{q}}{(a-x)^2}$

$(a-x)^2 = x^2$

$a - x = x$

$a = 2x$

$\boxed{x = \frac{a}{2}}$

∴ at the centre $(\frac{a}{2}, 0)$ of the line joining the two charge Electric field will be zero.

2) No, the net potential can't be zero for a system of two identical charges.

(b) (i) Consider two identical point charges located at points (0, 0) and (a, 0).

(1) Is there a point on the line joining them at which the electric field is zero ?

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Justify your answers for each case.

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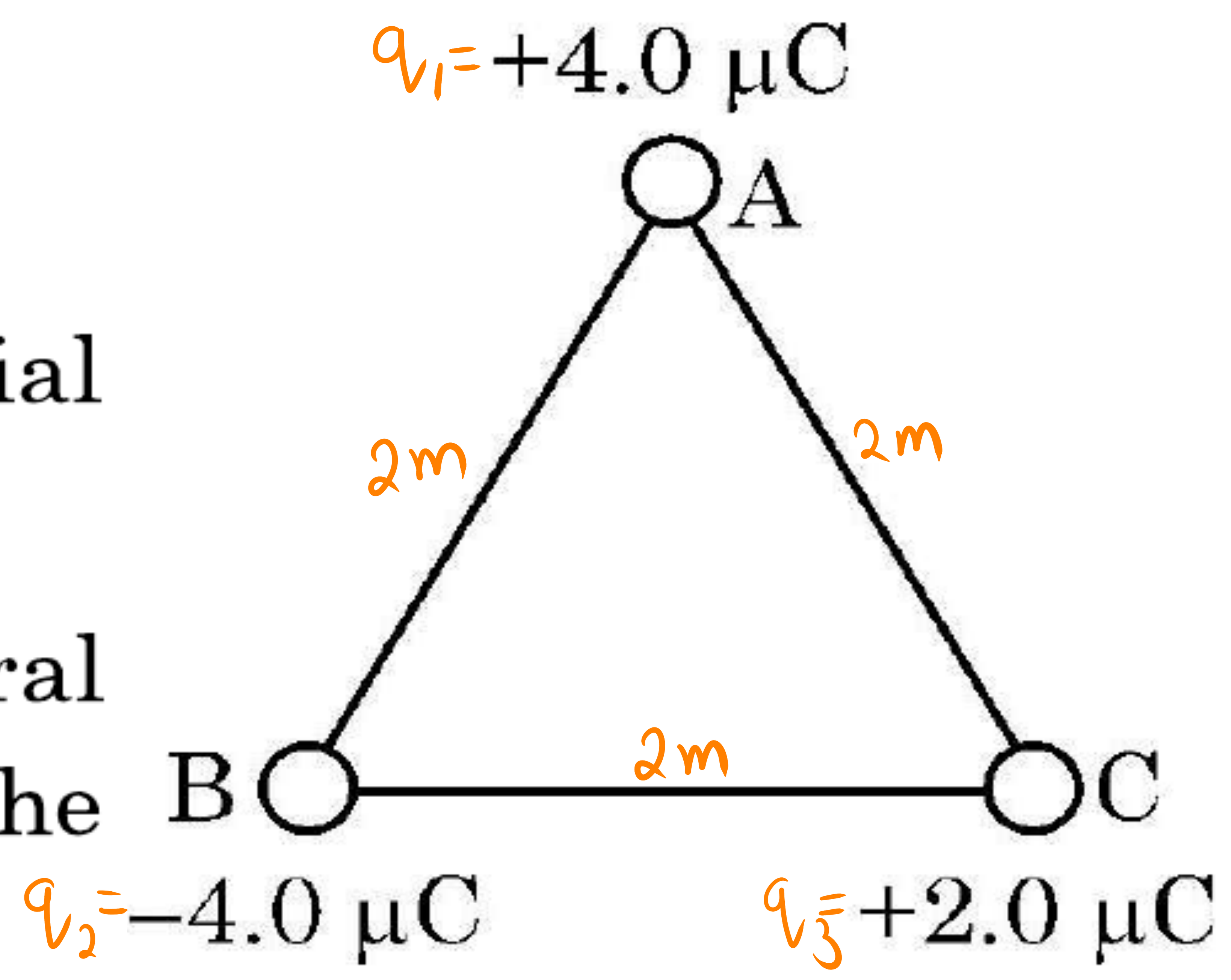
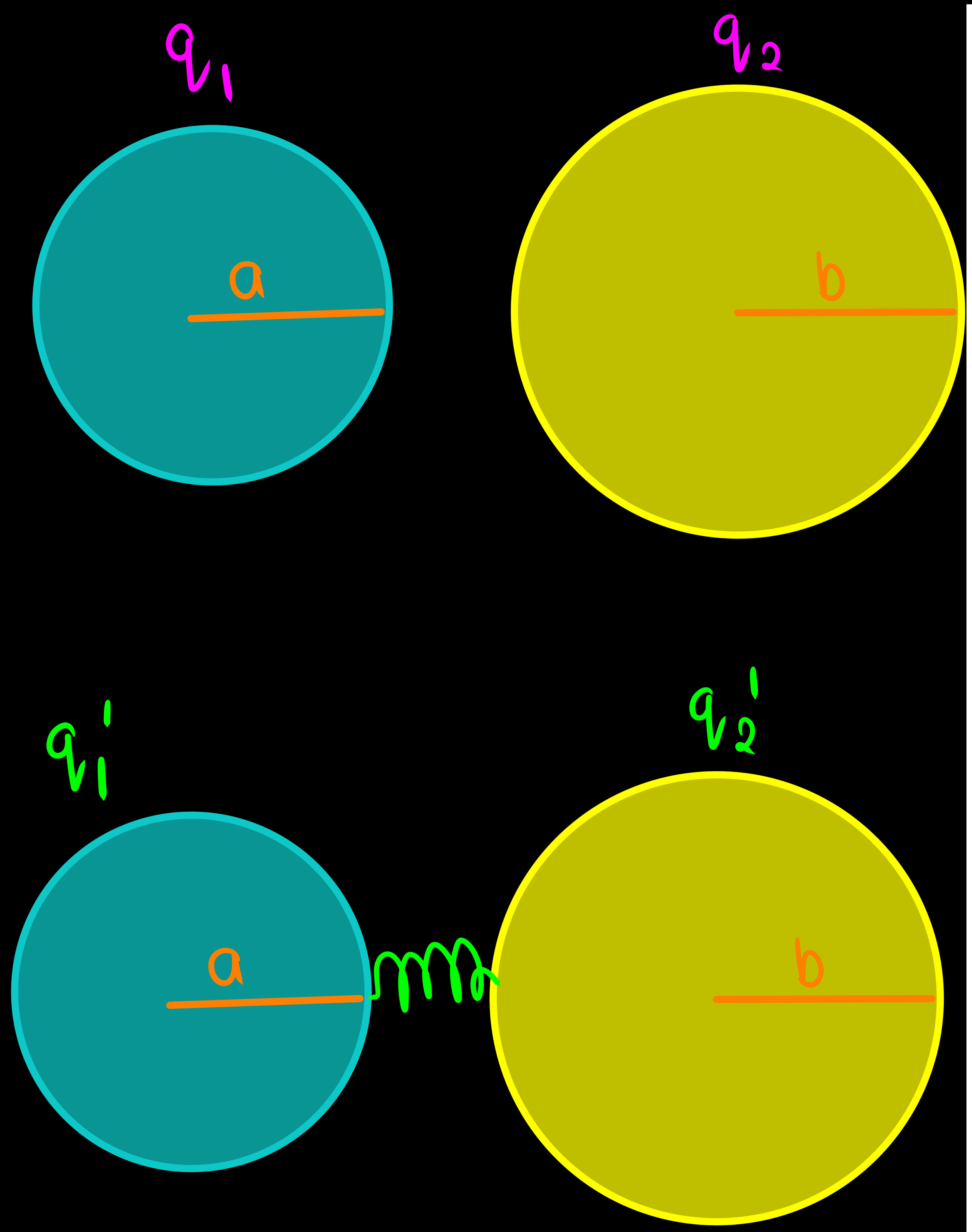




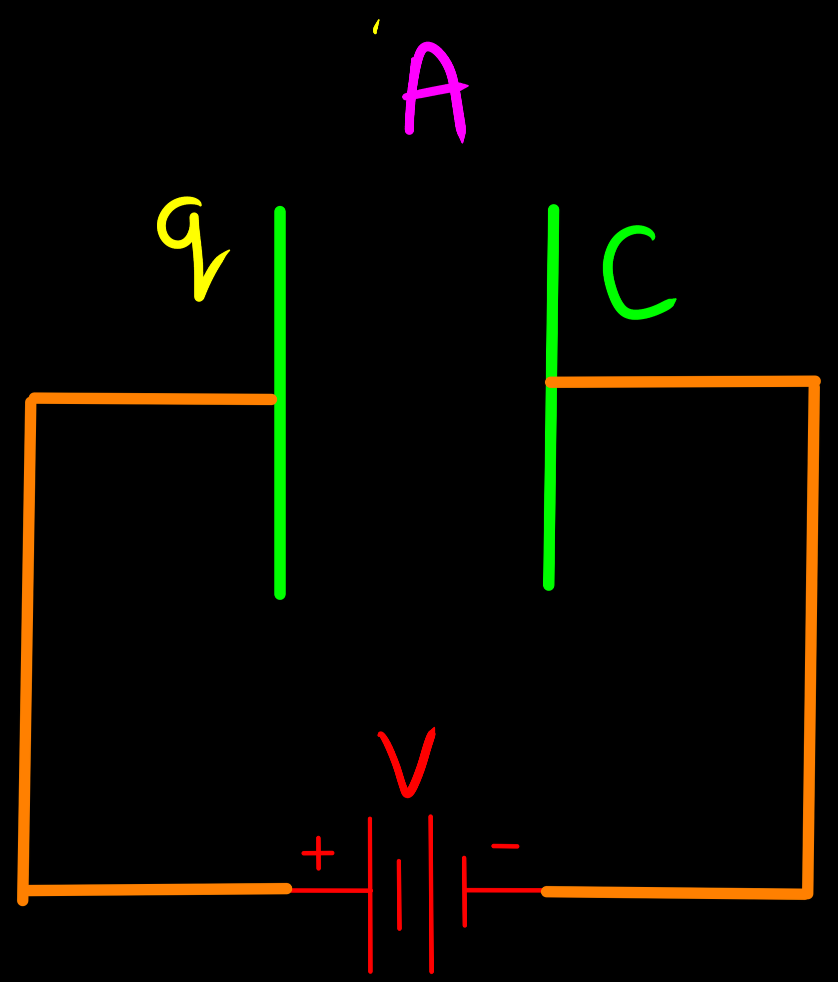
28. (a) Two charged conducting spheres of radii a and b are connected to each other by a wire. Find the ratio of the electric fields at their surfaces. **3**

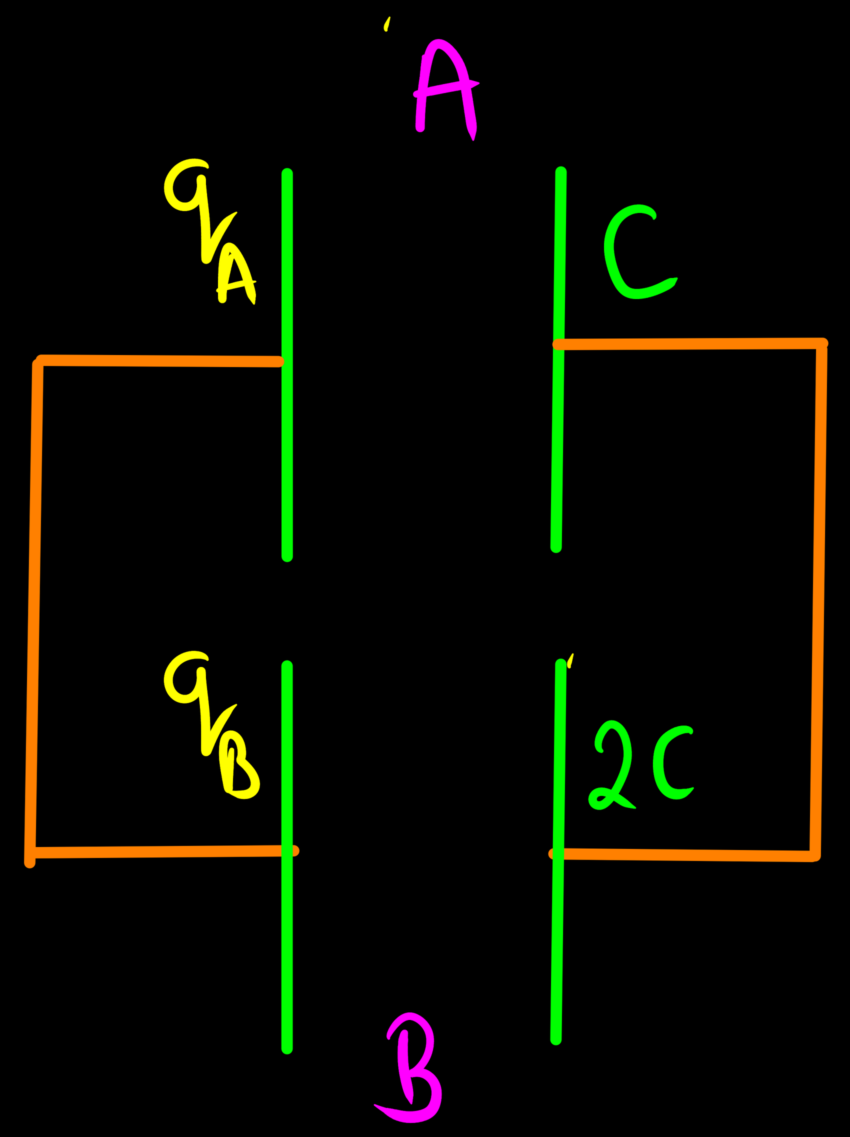
OR

(b) A parallel plate capacitor (A) of capacitance C is charged by a battery to voltage V. The battery is disconnected and an uncharged capacitor (B) of capacitance 2C is connected across A. Find the ratio of

(i) final charges on A and B.

(ii) total electrostatic energy stored in A and B finally and that stored in A initially.





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18. **Assertion (A) :** Work done in moving a charge around a closed path, in an electric field is always zero.

Reason (R) : Electrostatic force is a conservative force. **1**

12. The capacitors, each of 4 μF are to be connected in such a way that the effective capacitance of the combination is 6 μF. This can be achieved by connecting **1**

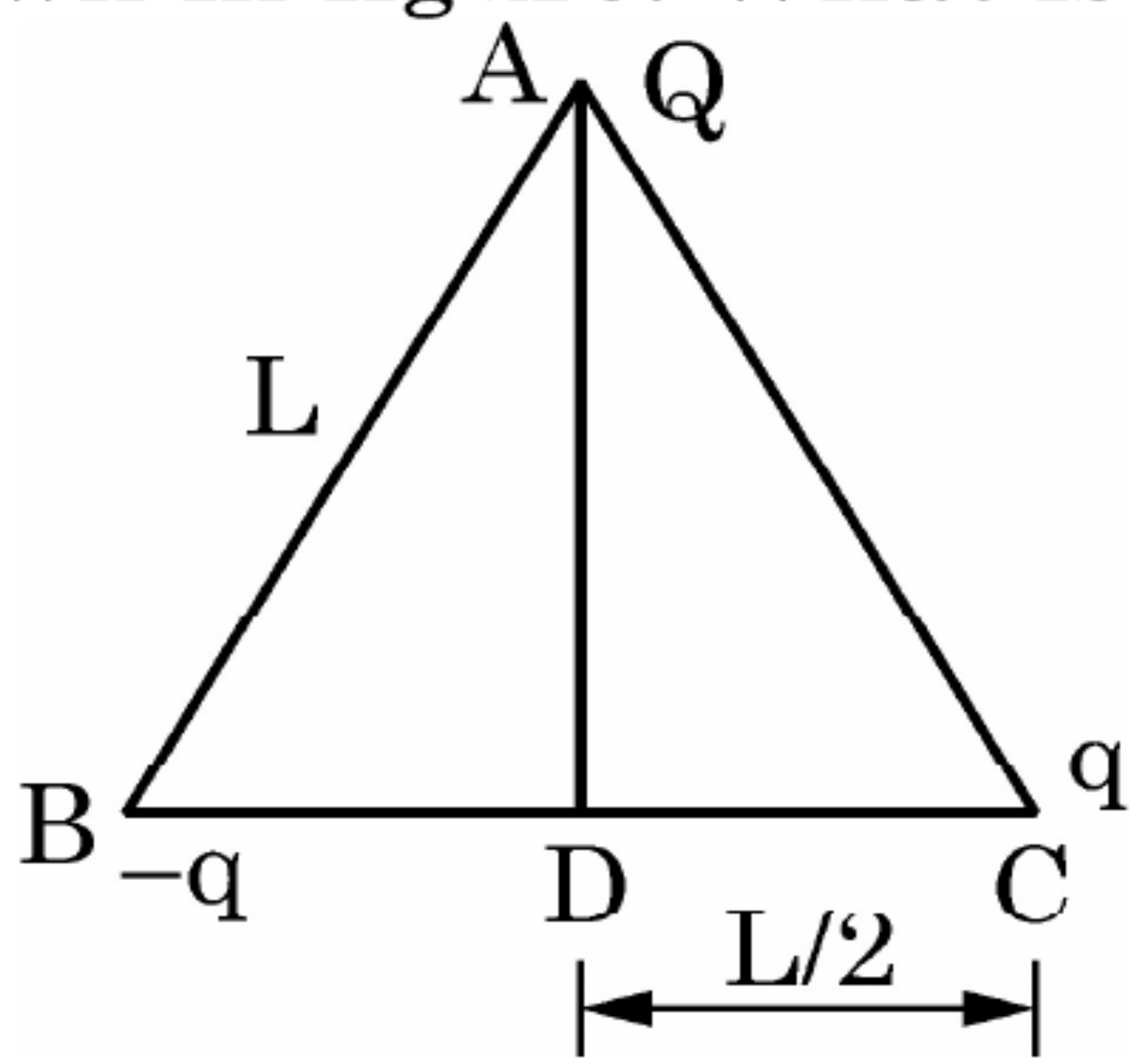
(A) All three in parallel

(B) All three in series

(C) Two of them connected in series and the combination in parallel to the third.

(D) Two of them connected in parallel and the combination in series to the third.

22. Three point charges Q, q and – q are kept at the vertices of an equilateral triangle of side L as shown in figure. What is **2**

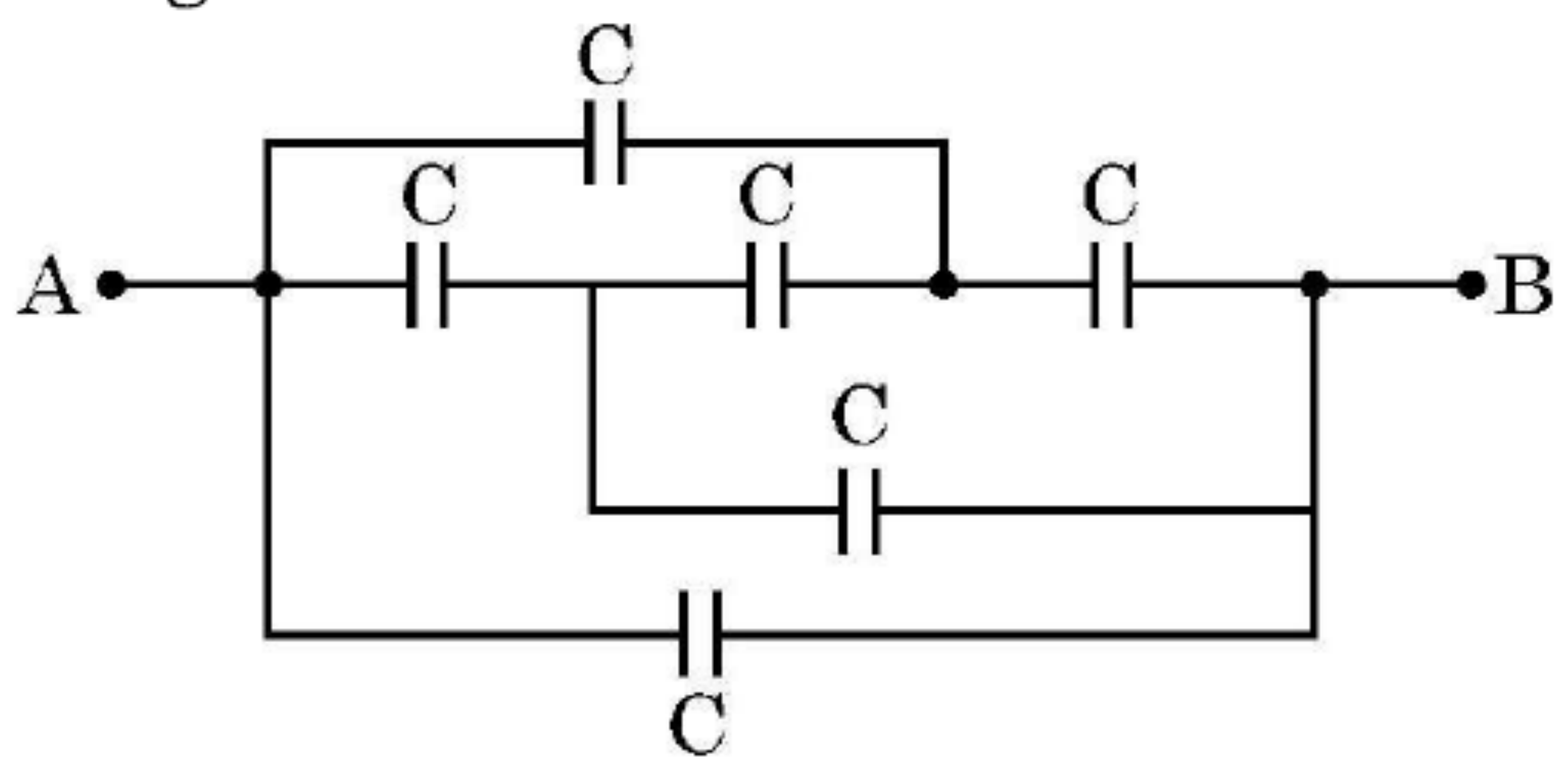


(i) the electrostatic potential energy of the arrangement ? and

(ii) the potential at point D ?

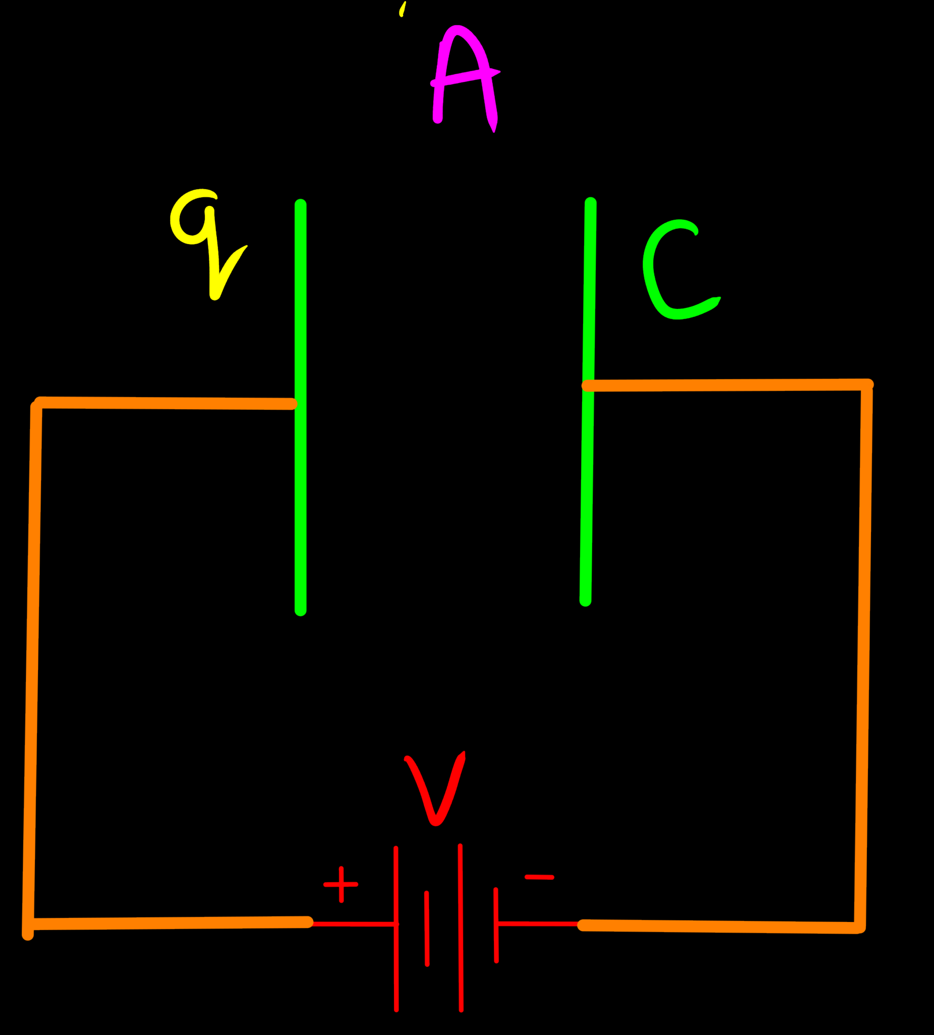
35. A capacitor is a system of two conductors separated by an insulator. The two conductors have equal and opposite charges with a potential difference between them. The capacitance of a capacitor depends on the geometrical configuration (shape, size and separation) of the system and also on the nature of the insulator separating the two conductors. They are used to store charges. Like resistors, capacitors can be arranged in series or parallel or a combination of both to obtain desired value of capacitance. **4**

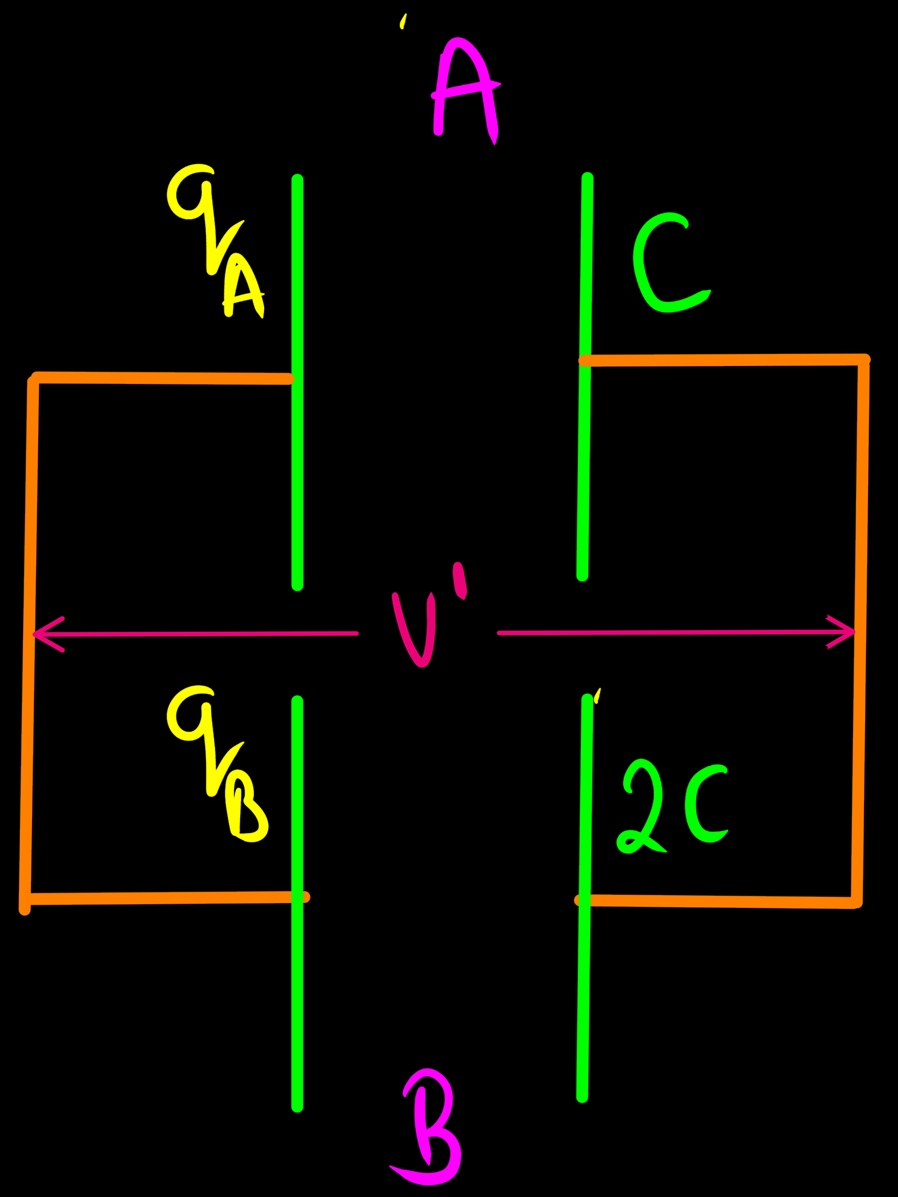
(i) Find the equivalent capacitance between points A and B in the given diagram.



(ii) A dielectric slab is inserted between the plates of a parallel plate capacitor. The electric field between the plates decreases. Explain.

(iii) A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.





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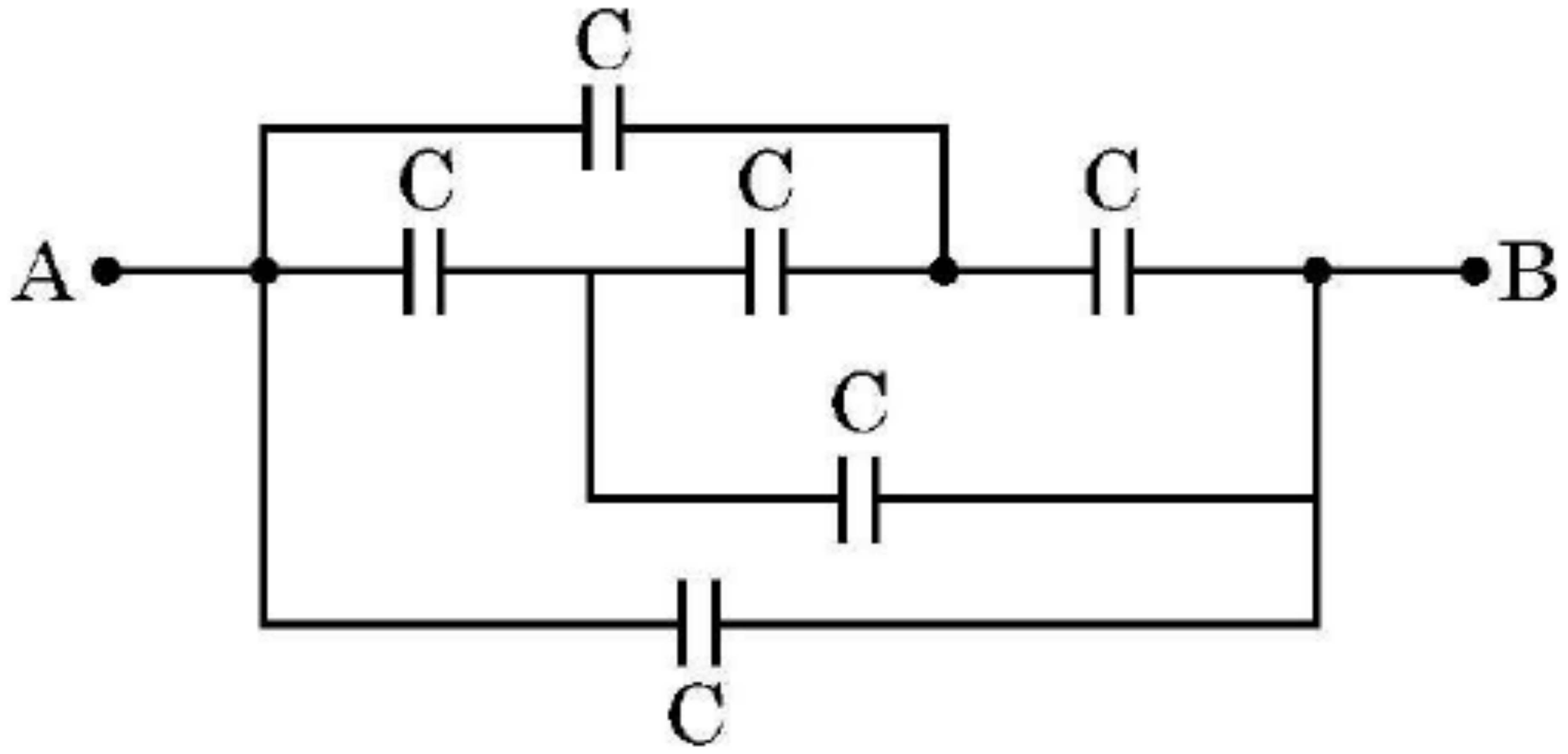

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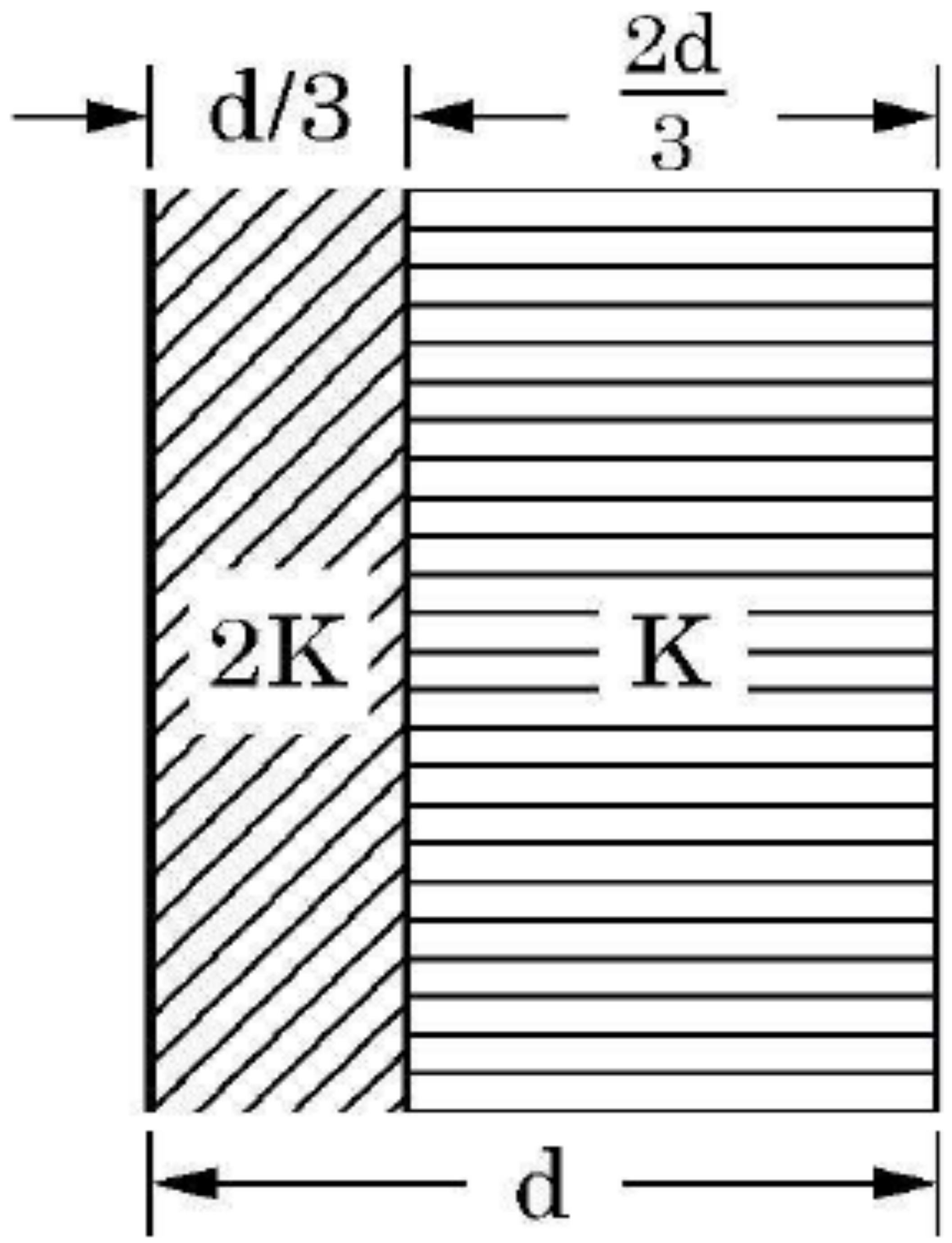
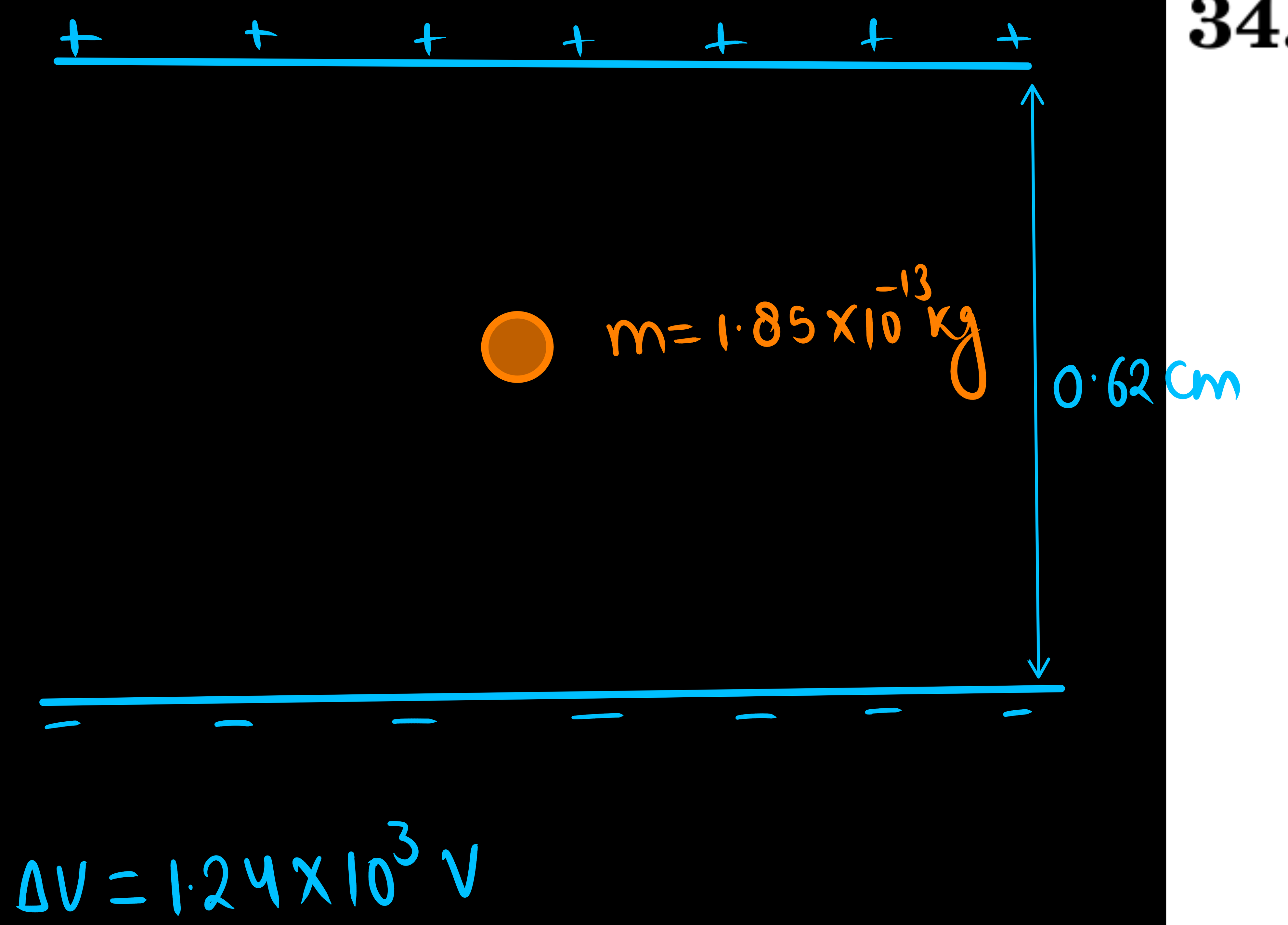
(iii) A capacitor A of capacitance C, having charge Q is connected across another uncharged capacitor B of capacitance 2C. Find an expression for (a) the potential difference across the combination and (b) the charge lost by capacitor A.

OR

(iii) Two slabs of dielectric constants 2K and K fill the space between the plates of a parallel plate capacitor of plate area A and plate separation d as shown in figure. Find an expression for capacitance of the system.



22. Depict the orientation of an electric dipole in (a) stable and (b) unstable equilibrium in an external uniform electric field.
Write the potential energy of the dipole in each case. **2**



34. Case Study : A charged body inside an electric field.

A charged latex sphere of mass $1{\cdot}85 \times 10^{-13}$ kg is held stationary in between two horizontal plates which are separated by a distance $0{\cdot}62$ cm. The potential difference between the plates is $1{\cdot}24 \times 10^{3}$ V with the upper plate being positive.

Based on the above facts, answer the following questions :

(i) What is the nature of charge on the latex sphere ? *1*

(ii) What is the direction of electric field between the plates ? *1*

(iii) What is the magnitude of electric field between the plates ? *2*

OR

(iii) What is the magnitude of charge on the latex sphere ? (Take $g = 10\ ms^{-2}$) *2*

(b) (i) How will the capacitance of a parallel plate capacitor change if :

(1) the plates area is doubled ?

(2) the separation between the plates is doubled ?

(ii) The effective capacitance of three capacitors of the same capacitance connected in series is 1 μF. Find the : *5*

(1) effective capacitance if they are connected in parallel.

(2) ratio of energy stored in the parallel combination of the capacitors to that in the series combination, if the combinations are connected to the same source one by one.

17. *Assertion (A) :* The electrostatic field $\vec{E}$ is a conservative field.

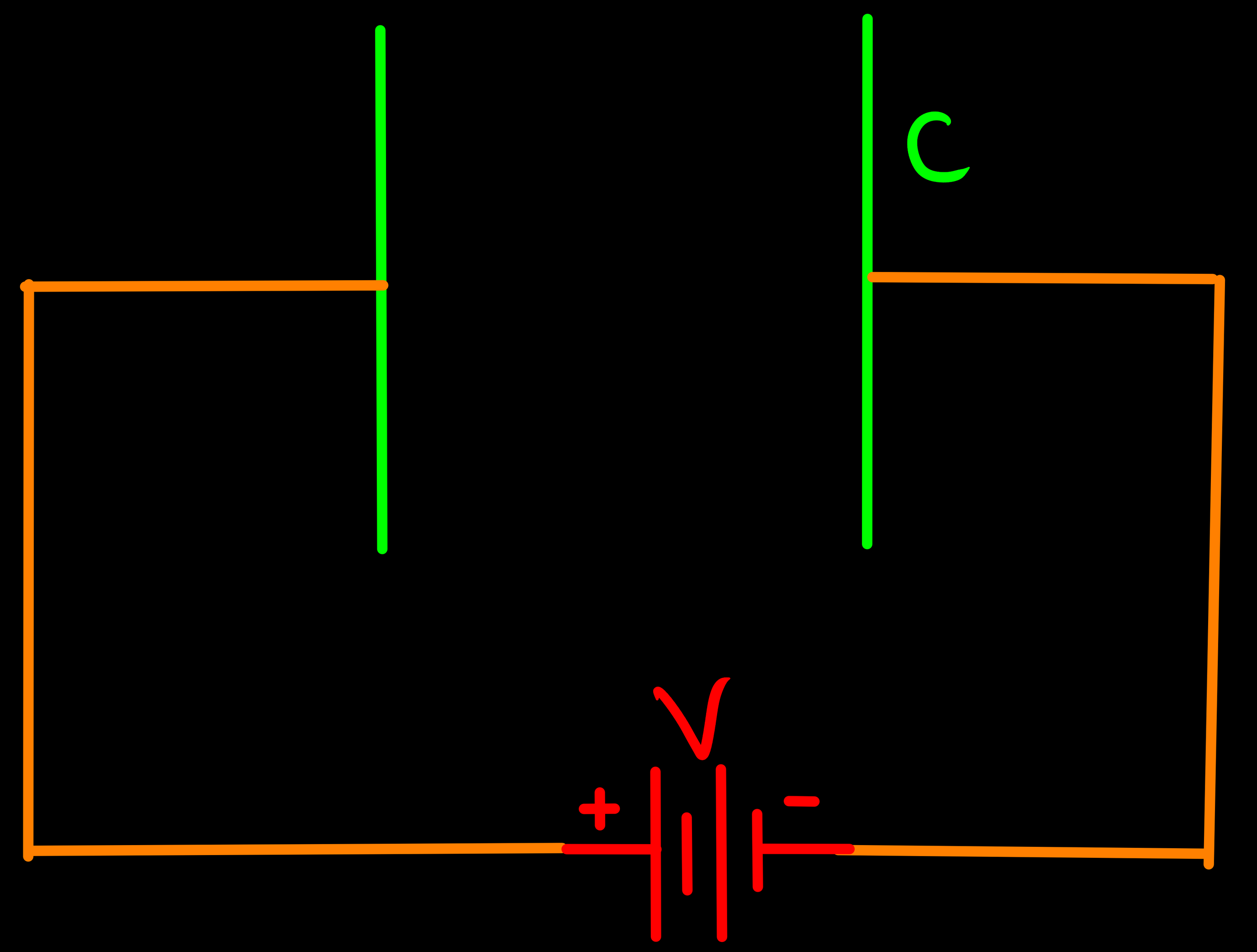
Reason (R) : Line integral of electric field $\vec{E}$ around a closed path is non zero.

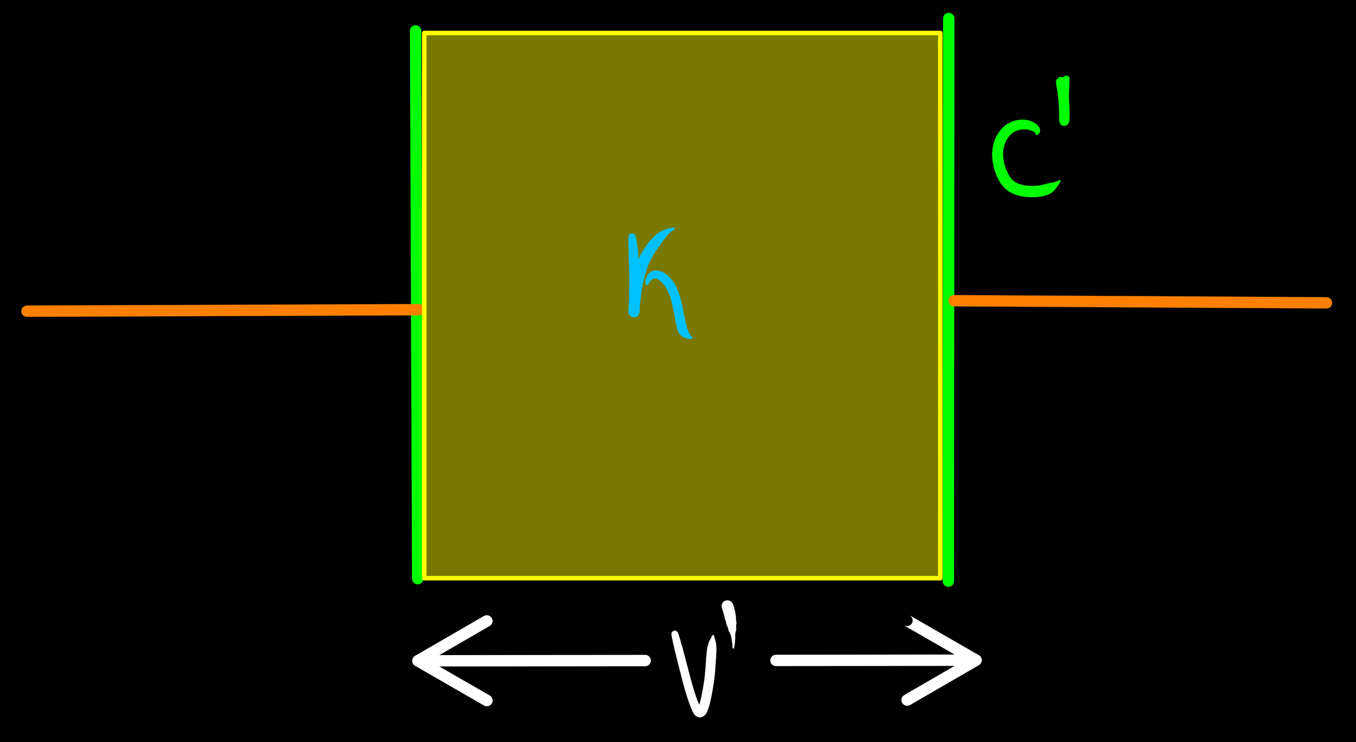
A parallel plate capacitor is an arrangement of two identical metal plates kept parallel, a small distance apart. The capacitance of a capacitor depends on the size and separation of the two plates and also on the dielectric constant of the medium between the plates. Like resistors, capacitors can also be arranged in series or parallel or a combination of both. By virtue of electric field between the plates, charged capacitors store energy.

(a) The capacitance of a parallel plate capacitor increases from 10 μF to 80 μF on introducing a dielectric medium between the plates. Find the dielectric constant of the medium. *1*

(b) n capacitors, each of capacitance C, are connected in series. Find the equivalent capacitance of the combination. *1*

(c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer. *2* c)





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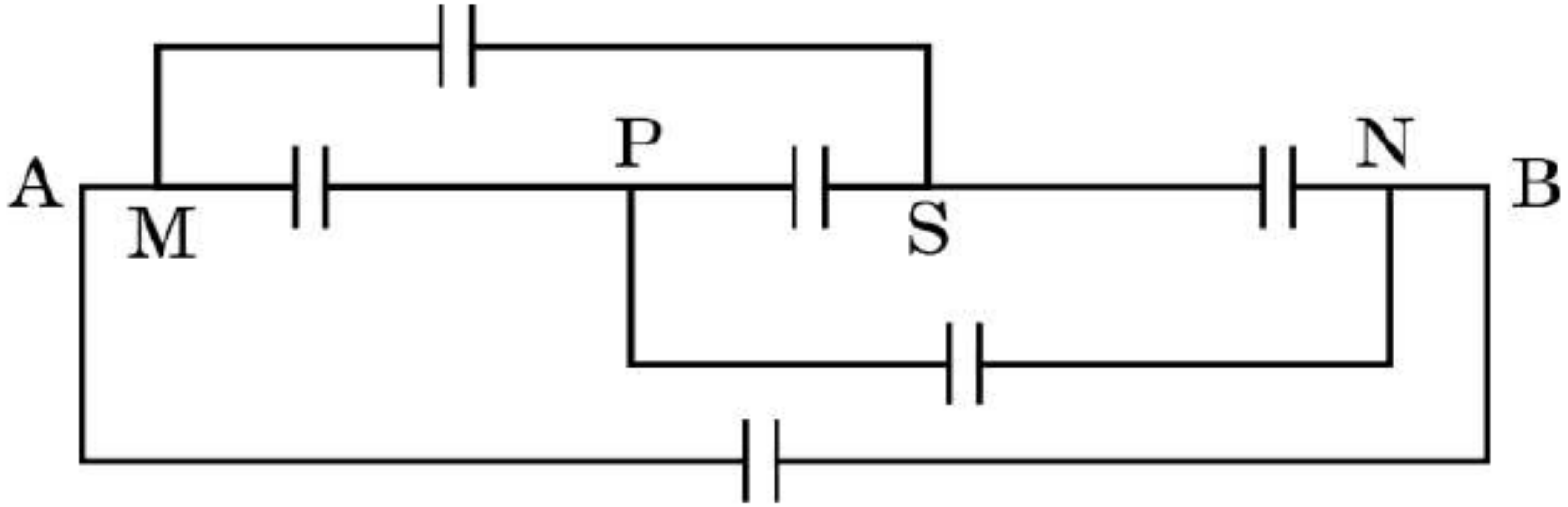
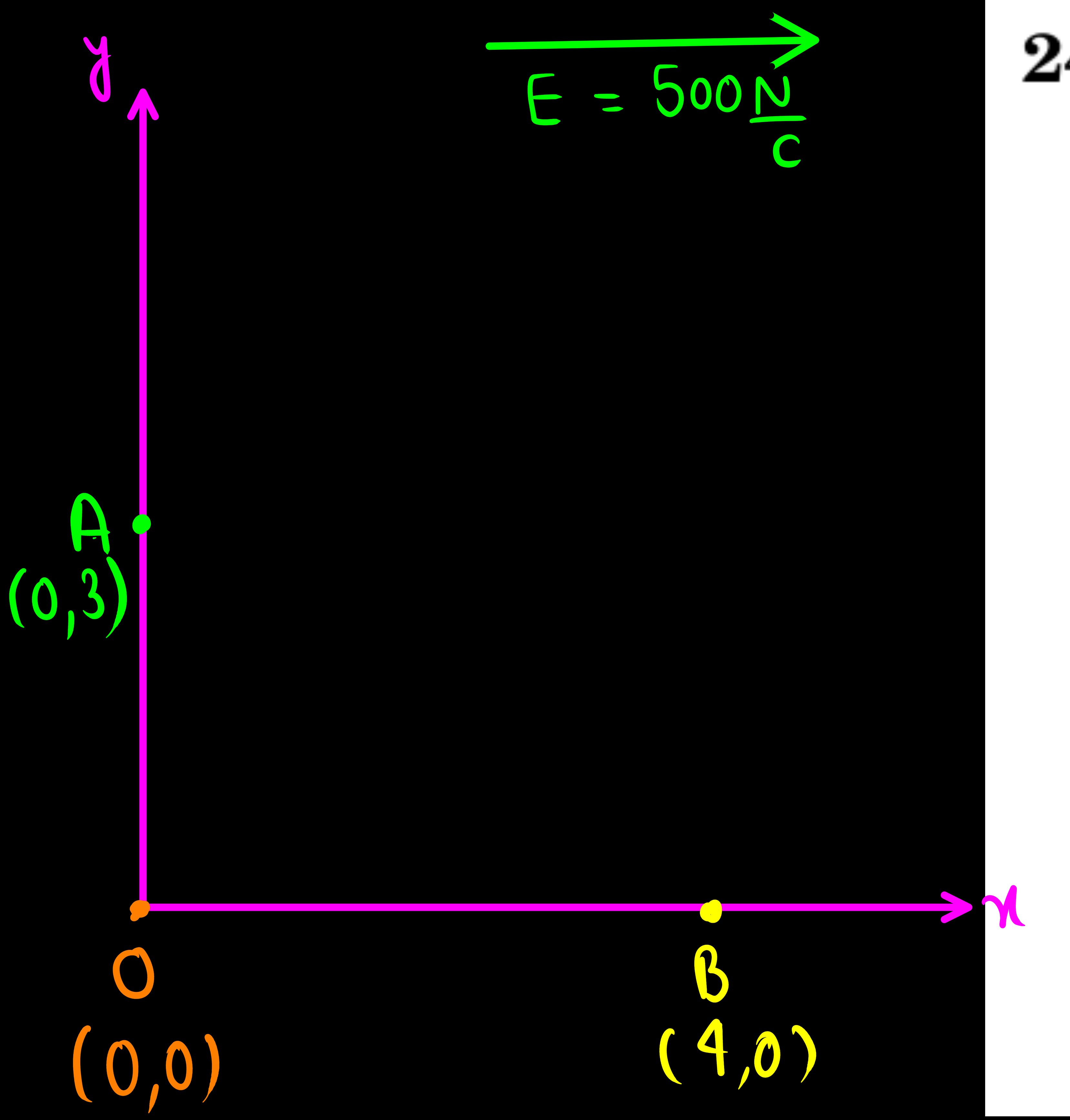
(c) A capacitor is charged to a potential (V) by connecting it to a battery. After some time, the battery is disconnected and a dielectric is introduced between the plates. How will the potential difference between the plates, and the energy stored in it be affected ? Justify your answer. *2*

(c) Find the equivalent capacitance between points A and B, if capacitance of each capacitor is C. 2

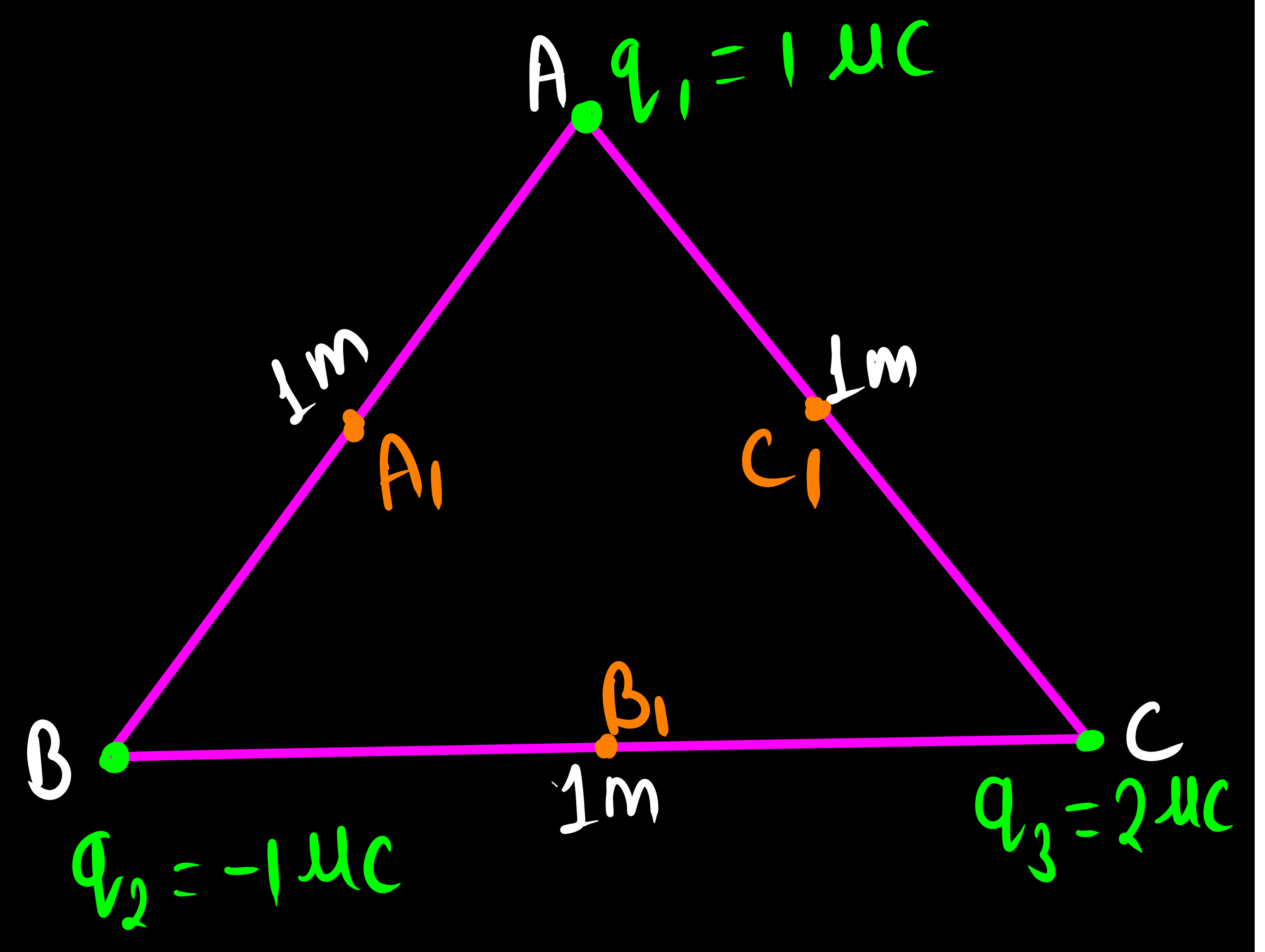




24. (a) A uniform electric field E of 500 N/C is directed along +x axis. O, B and A are three points in the field having x and y coordinates (in cm) (0, 0), (4, 0) and (0, 3) respectively. Calculate the potential difference between the points (i) O and A, and (ii) O and B. *2*

OR

(b) Three point charges 1 μC, –1 μC and 2 μC are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. *2*



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OR

(b) Three point charges $1\ \mu C$, $-1\ \mu C$ and $2\ \mu C$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1 and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. *2*

33. (a) (i) Draw equipotential surfaces for an electric dipole.

(ii) Two point charges q_1 and q_2 are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in an external electric field $\vec{E}$. Obtain an expression for the potential energy of the system.

(iii) The dipole moment of a molecule is 10^{-30} Cm. It is placed in an electric field $\vec{E}$ of 10^5 V/m such that its axis is along the electric field. The direction of $\vec{E}$ is suddenly changed by 60° at an instant. Find the change in the potential energy of the dipole, at that instant.

30. Dielectrics play an important role in design of capacitors. The molecules of a dielectric may be polar or non-polar. When a dielectric slab is placed in an external electric field, opposite charges appear on the two surfaces of the slab perpendicular to electric field. Due to this an electric field is established inside the dielectric.

The capacitance of a capacitor is determined by the dielectric constant of the material that fills the space between the plates. Consequently, the energy storage capacity of a capacitor is also affected. Like resistors, capacitors can also be arranged in series and/or parallel.

(i) Which of the following is a polar molecule ?

(A) O_2 (B) H_2

(C) N_2 (D) HCl

(ii) Which of the following statements about dielectrics is correct ?

(A) A polar dielectric has a net dipole moment in absence of an external electric field which gets modified due to the induced dipoles.

(B) The net dipole moments of induced dipoles is along the direction of the applied electric field.

(C) Dielectrics contain free charges.

(D) The electric field produced due to induced surface charges inside a dielectric is along the external electric field.

(iii) When a dielectric slab is inserted between the plates of an isolated charged capacitor, the energy stored in it :

(A) increases and the electric field inside it also increases.

(B) decreases and the electric field also decreases.

(C) decreases and the electric field increases.

(D) increases and the electric field decreases.

(iv) (a) An air-filled capacitor with plate area A and plate separation d has capacitance C_0. A slab of dielectric constant K, area A and thickness $\left(\frac{d}{5}\right)$ is inserted between the plates. The capacitance of the capacitor will become

(A) $\left[\frac{4K}{5K+1}\right]C_0$ (B) $\left[\frac{K+5}{4}\right]C_0$

(C) $\left[\frac{5K}{4K+1}\right]C_0$ (D) $\left[\frac{K+4}{5K}\right]C_0$

OR

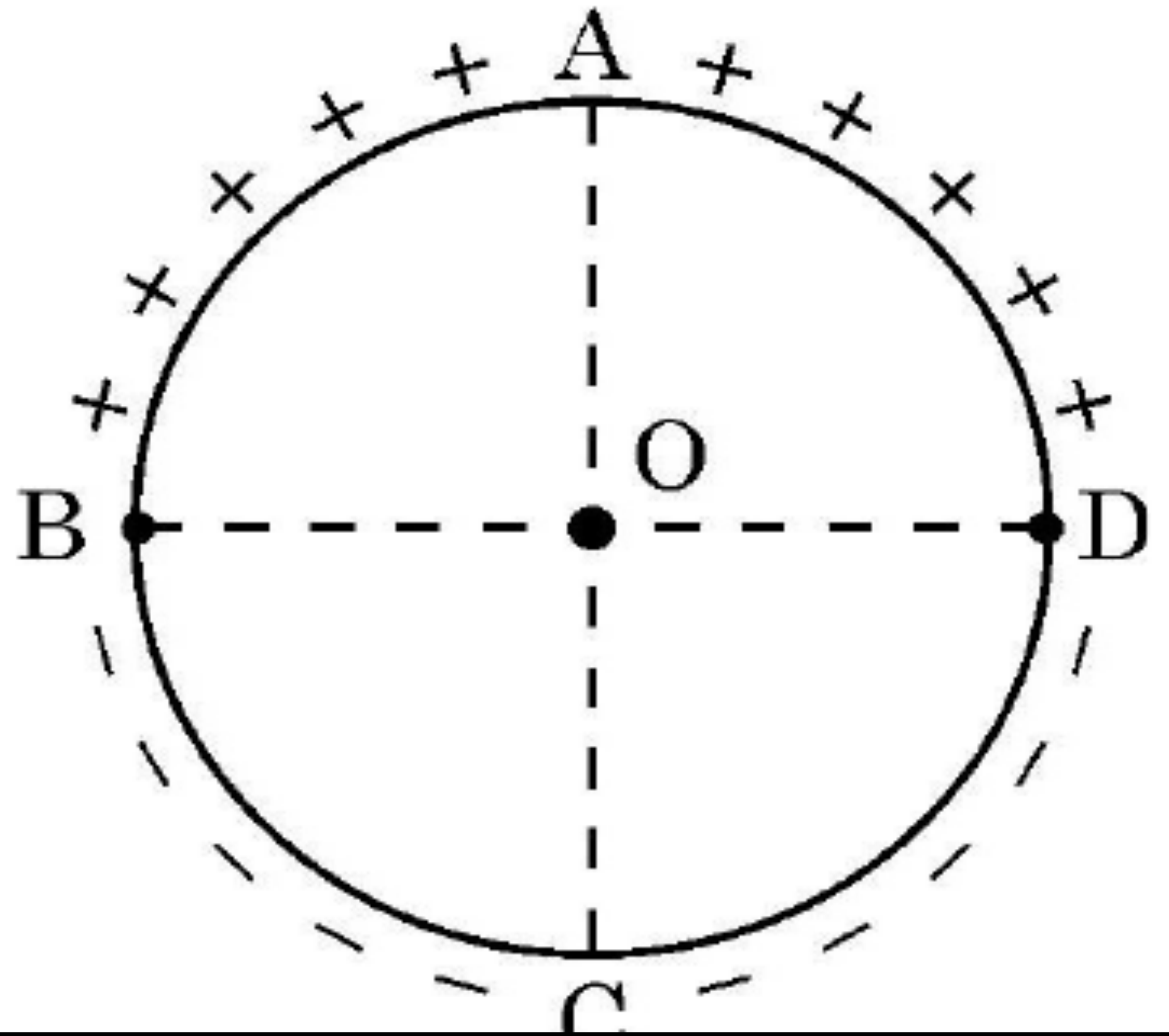
(iv) (b) Two capacitors of capacitances $2\ C_0$ and $6\ C_0$ are first connected in series and then in parallel across the same battery. The ratio of energies stored in series combination to that in parallel is

(A) $\frac{1}{4}$ (B) $\frac{1}{6}$

(C) $\frac{2}{15}$ (D) $\frac{3}{16}$

15. **Assertion (A) :** Equal amount of positive and negative charges are distributed uniformly on two halves of a thin circular ring as shown in figure. The resultant electric field at the centre O of the ring is along OC.

Reason (R) : It is so because the net potential at O is not zero.



31. (a) (i) A dielectric slab of dielectric constant 'K' and thickness 't' is inserted between plates of a parallel plate capacitor of plate separation d and plate area A. Obtain an expression for its capacitance.

(ii) Two capacitors of different capacitances are connected first (1) in series and then (2) in parallel across a dc source of 100 V. If the total energy stored in the combination in the two cases are 40 mJ and 250 mJ respectively, find the capacitance of the capacitors. **5**

23. Two conducting spherical shells A and B of radii R and 2R are kept far apart and charged to the same charge density σ. They are connected by a wire. Obtain an expression for final potential of shell A. **3**

1. An electric dipole of dipole moment $\vec{p}$ is kept in a uniform electric field $\vec{E}$. The amount of work done to rotate it from the position of stable equilibrium to that of unstable equilibrium will be **1**

(A) 2 pE (B) –2 pE

(C) pE (D) zero

23. A thin spherical conducting shell of radius R has a charge q. A point charge Q is placed at the centre of the shell. Find (i) The charge density on the outer surface of the shell and (ii) the potential at a distance of (R/2) from the centre of the shell. **3**

1. Three point charges, each of charge q are placed on vertices of a triangle ABC, with AB = AC = 5L, BC = 6L. The electrostatic potential at midpoint of side BC will be **1**

(A) $\frac{11}{48}\frac{q}{\pi\varepsilon_0 L}$

(B) $\frac{8q}{36\pi\varepsilon_0 L}$

(C) $\frac{5q}{24\pi\varepsilon_0 L}$

(D) $\frac{1}{16}\frac{q}{\pi\varepsilon_0 L}$